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New representations of the Hu–Meyer formulas and series expansions of iterated Stratonovich stochastic integrals with respect to components of a multidimensional Wiener process

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Abstract. The article is devoted to the systematic derivation of new representations of the Hu–Meyer formulas. The formula expressing a multiple Wiener stochastic integral through the sum of multiple Stratonovich stochastic integrals and the formula expressing a multiple Stratonovich stochastic integral through the sum of multiple Wiener stochastic integrals are derived for the case of a multidimensional Wiener process. At that several different definitions of the multiple Stratonovich stochastic integral and several variants of sufficient conditions for the validity of the Hu–Meyer formulas are used. In particular, the proof method proposed by the author in 2006 is applied to obtain Hu–Meyers formulas based on generalized multiple Fourier series for the case of a multidimensional Wiener process. Of great importance for the numerical solution of Itô and Stratonovich stochastic differential equations is the verification of sufficient conditions for the applicability of the Hu–Meyer formula (based on generalized multiple Fourier series) for the case of iterated Stratonovich stochastic integrals with respect to components of a multidimensional Wiener process. In

the author's previous works, the indicated conditions were verified for iterated Stratonovich stochastic integrals of multiplicities 1 to 6 (the case of an arbitrary basis in the Hilbert space) and for iterated Stratonovich stochastic integrals of multiplicities 7 and 8 (the case of two special bases in Hilbert space (the trigonometric Fourier basis and the basis of Legendre polynomials)). Therefore, the results of the article will be usefull for constructing high-order strong numerical methods for non-commutative systems of Itô stochastic differential equations.
Key words: expansion, generalized multiple Fourier series, Hu–Meyer formula, iterated Stratonovich stochastic integral, iterated Itô stochastic integral, Itô stochastic differential equation, Legendre polynomial, mean-square convergence, multidimensional Wiener process, multiple Stratonovich stochastic integral, multiple Wiener stochastic integral.

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1 Introduction

In the theory of stochastic integrals, a special place is occupied by the so-called Hu–Meyer formulas, which express a multiple Wiener stochastic integral

through the sum of multiple Stratonovich stochastic integrals and, conversely, express a multiple Stratonovich stochastic integral through the sum of multiple Wiener stochastic integrals. Usually, the indicated multiple stochastic integrals in the Hu–Meier formulas are considered with respect to a scalar Wiener or Poisson processes [1]-[6]. It is worth noting the Hu–Meyer formula for multiple stochastic integrals with respect to Lévy processes [7].

Of great interest (in the context of numerical integration of Itô stochastic differential equations (SDEs)) is the Hu–Meyer formula, which express the iterated Itô stochastic integral with respect to the components of a multidimensional Wiener process through the sum of multiple Stratonovich stochastic integrals. A version of such a formula based on generalized multiple Fourier series was first obtained in 2006 [8] (also see [9]-[33]). The inverse version of the above formula, which expresses the multiple Stratonovich stochastic integral in terms of multiple Wiener stochastic integrals, can be found in [34].

Rest of the work is organized as follows. In Sect. 2.1 we give a brief review of some old results on the expansion of iterated Stratonovich and Itô stochastic integrals obtained by the author. Sect. 3.1 and 3.2 are devoted to the systematic derivation of new representations for both of the indicated variants of the Hu–Meyer formula: the formula expressing a multiple Wiener stochastic integral through the sum of multiple Stratonovich stochastic integrals, and the formula expressing a multiple Stratonovich stochastic integral through the sum of multiple Wiener stochastic integrals (the case of a multidimensional Wiener process). Several different definitions of the multiple Stratonovich stochastic integral and several variants of sufficient conditions for the validity of the Hu–Meyer formulas are used. At that, we use the proof method from our works [8], [18], [19] to obtain Hu–Meyers formulas based on genealized multiple Fourier series for the case of a multidimensional Wiener process.

It should be noted that of great importance for the numerical integration of Itô SDEs is the verification of sufficient conditions for the applicability of the Hu–Meyer formula (based on generalized multiple Fourier series) for the case of iterated Stratonovich stochastic integrals with respect to components of a multidimensional Wiener process. In the author's previous works [13]-[17] the indicated conditions were verified for iterated Stratonovich stochastic integrals of multiplicities 1 to 6 (the case of an arbitrary basis in the Hilbert space) and for iterated Stratonovich stochastic integrals of multiplicities 7 and 8 (the case of two special bases in Hilbert space (the trigonometric Fourier basis and the basis of Legendre polynomials)).

2 Preliminary Results

2.1 A Review of Some Earlier Results on the Expansion of Iterated Stratonovich and Itô Stochastic Integrals

In this section, we will give a brief review of some earlier results on the expansion of iterated Stratonovich and Itô stochastic integrals obtained by the author (the case of an arbitrary complete orthonormal system (CONS) in $L_2([t,T])$).

Let $(\Omega, \mathrm{F}, \mathsf{P})$ be a complete probability space, let $\{\mathrm{F}_t, t \in [0,T]\}$ be a nondecreasing right-continous family of σ-subalgebras of F, and let $\mathbf{w}_t = (\mathbf{w}_t^{(1)} \ldots \mathbf{w}_t^{(m)})^{\mathsf{T}}$ be a standard m-dimensional Wiener stochastic process with independent components which is F_t-measurable for any $t \in [0,T]$. Consider the following iterated Stratonovich stochastic integrals:

$$J^*[\psi^{(k)}]_{T,t}^{(i_1\ldots i_k)} = \int\limits_t^{*T} \psi_k(t_k) \ldots \int\limits_t^{*t_2} \psi_1(t_1) d\mathbf{w}_{t_1}^{(i_1)} \ldots d\mathbf{w}_{t_k}^{(i_k)}, \tag{1}$$

where $\psi_1(\tau), \ldots, \psi_k(\tau) : [t,T] \to \mathbb{R}$, $i_1, \ldots, i_k = 0, 1, \ldots, m$, $\mathbf{w}_\tau^{(0)} = \tau$.

Note that the importance of the problem of numerical integration of Itô SDEs is explained by a wide range of their applications [35]-[37]. One of the effective approaches to the above problem is based on the so-called Taylor–Stratonovich expansions [35] (also see [16], Chapter 4). The stochastic integrals (1) are multipliers for terms of the mentioned expansions that explains the relevance of the topic of the work.

Throughout this article we will use the definition of the iterated Stratonovich stochastic integral from [35] (also see [16], Sect. 2.1.1).

Let $\psi_1(\tau), \ldots, \psi_k(\tau) \in L_2([t,T])$. Consider the Fourier coefficient

$$C_{j_k\ldots j_1} = \int\limits_t^T \psi_k(t_k)\phi_{j_k}(t_k) \ldots \int\limits_t^{t_2} \psi_1(t_1)\phi_{j_1}(t_1) dt_1 \ldots dt_k \tag{2}$$

corresponding to the factorized Volterra–type kernel of the form

$$K(t_1, \ldots, t_k) = \begin{cases} \psi_1(t_1) \ldots \psi_k(t_k), & t_1 < \ldots < t_k \\ 0, & \text{otherwise} \end{cases}, \tag{3}$$

where $(t_1,\ldots,t_k)\in[t,T]^k$ $(k\ge 2)$ and $\{\phi_j(x)\}_{j=0}^{\infty}$ is an arbitrary CONS in $L_2([t,T])$.

Theorem 1 [16] (Sect. 2.1.4). *Suppose that $\{\phi_j(x)\}_{j=0}^{\infty}$ is an arbitrary CONS in $L_2([t,T])$. Moreover, $\psi_1(\tau)$ and $\psi_2(\tau)$ are continuous functions on $[t,T]$. Then*

$$J^*[\psi^{(2)}]_{T,t}^{(i_1i_2)}=\underset{p_1,p_2\to\infty}{\text{l.i.m.}}\sum_{j_1=0}^{p_1}\sum_{j_2=0}^{p_2}C_{j_2j_1}\zeta_{j_1}^{(i_1)}\zeta_{j_2}^{(i_2)}, \tag{4}$$

where $J^*[\psi^{(2)}]_{T,t}^{(i_1i_2)}$ is defined by (1), $i_1,i_2=0,1,\ldots,m$, l.i.m. *is a limit in the mean-square sense, $C_{j_2j_1}$ has the form (2),*

$$\zeta_j^{(i)}=\int\limits_t^T\phi_j(\tau)d\mathbf{w}_\tau^{(i)}$$

are independent standard Gaussian random variables for various i or j (in the case $i\ne 0$), and $\mathbf{w}_\tau^{(0)}=\tau$.

Theorem 2 [14], [16] (Sect. 2.24, 2.33). *Suppose that $\{\phi_j(x)\}_{j=0}^{\infty}$ is an arbitrary CONS in $L_2([t,T])$ and $\psi_i(\tau)=(\tau-t)^{l_i}$ $(l_i=0,1,2,\ldots,\ i=1,\ldots,4)$. Then*

$$J^*[\psi^{(k)}]_{T,t}^{(i_1\ldots i_k)}=\underset{p\to\infty}{\text{l.i.m.}}\sum_{j_1,\ldots,j_k=0}^{p}C_{j_k\ldots j_1}\zeta_{j_1}^{(i_1)}\ldots\zeta_{j_k}^{(i_k)}\quad(k=3,\ 4),$$

where $J^*[\psi^{(k)}]_{T,t}^{(i_1\ldots i_k)}$ $(k=3,\ 4)$ is defined by (1), $i_1,\ldots,i_4=0,1,\ldots,m$, $C_{j_k\ldots j_1}$ has the form (2); another notations as in Theorem 1.

Theorem 3 [14], [16] (Sect. 2.32, 2.34). *Suppose that $\{\phi_j(x)\}_{j=0}^{\infty}$ is an arbitrary CONS in $L_2([t,T])$ and $\psi_1(\tau),\ldots,\psi_6(\tau)\equiv 1$. Then*

$$J^*[\psi^{(k)}]_{T,t}^{(i_1\ldots i_k)}=\underset{p\to\infty}{\text{l.i.m.}}\sum_{j_1,\ldots,j_k=0}^{p}C_{j_k\ldots j_1}\zeta_{j_1}^{(i_1)}\ldots\zeta_{j_k}^{(i_k)}\quad(k=5,\ 6),$$

where $J^*[\psi^{(k)}]_{T,t}^{(i_1\ldots i_k)}$ $(k=5,\ 6)$ is defined by (1), $i_1,\ldots,i_6=0,1,\ldots,m$, $C_{j_k\ldots j_1}$ has the form (2); another notations as in Theorem 1.

Consider the unordered set $\{1,2,\ldots,k\}$ and separate it into two parts: the first part consists of r unordered pairs (sequence order of these pairs is also

unimportant) and the second one consists of the remaining $k-2r$ numbers. So, we have

$$(\{\underbrace{\{g_1,g_2\},\ldots,\{g_{2r-1},g_{2r}\}}_{\text{part 1}}\},\{\underbrace{q_1,\ldots,q_{k-2r}}_{\text{part 2}}\}), \tag{5}$$

where $\{g_1,g_2,\ldots,g_{2r-1},g_{2r},q_1,\ldots,q_{k-2r}\}=\{1,2,\ldots,k\}$, braces mean an unordered set, and parentheses mean an ordered set.

Denote

$$C_{j_k\ldots j_{l+1}j_lj_{l-1}j_{l-2}\ldots j_1}\Big|_{(j_lj_{l-1})\curvearrowright(\cdot)} \stackrel{\text{def}}{=}$$

$$\stackrel{\text{def}}{=}\int\limits_t^T\psi_k(t_k)\phi_{j_k}(t_k)\ldots\int\limits_t^{t_{l+2}}\psi_{l+1}(t_{l+1})\phi_{j_{l+1}}(t_{l+1})\int\limits_t^{t_{l+1}}\psi_l(t_l)\psi_{l-1}(t_l)\times$$

$$\times\int\limits_t^{t_l}\psi_{l-2}(t_{l-2})\phi_{j_{l-2}}(t_{l-2})\ldots\int\limits_t^{t_2}\psi_1(t_1)\phi_{j_1}(t_1)dt_1\ldots dt_{l-2}dt_lt_{l+1}\ldots dt_k, \tag{6}$$

where $\{l,l-1\}$ is one of the pairs $\{g_1,g_2\},\ldots,\{g_{2r-1},g_{2r}\}$ (see (5)).

Further, we will use the quantity

$$\prod_{l=1}^r\mathbf{1}_{\{g_{2l}=g_{2l-1}+1\}}C_{j_k\ldots j_1}\Big|_{(j_{g_2}j_{g_1})\curvearrowright(\cdot)\ldots(j_{g_{2r}}j_{g_{2r-1}})\curvearrowright(\cdot),j_{g_1}=j_{g_2},\ldots,j_{g_{2r-1}}=j_{g_{2r}}}$$

which is defined iteratively using the equality (6), where $\mathbf{1}_A$ is the indicator of the set A.

Introduce the following notations:

$$J\left[\psi^{(k)}\right]_{T,t}^{(i_1\ldots i_k)[s_l,\ldots,s_1]} \stackrel{\text{def}}{=} \prod_{q=1}^l\mathbf{1}_{\{i_{s_q}=i_{s_q+1}\neq0\}}\times$$

$$\times\int\limits_t^T\psi_k(t_k)\ldots\int\limits_t^{t_{s_l+3}}\psi_{s_l+2}(t_{s_l+2})\int\limits_t^{t_{s_l+2}}\psi_{s_l}(t_{s_l+1})\psi_{s_l+1}(t_{s_l+1})\times$$

$$\times\int\limits_t^{t_{s_l+1}}\psi_{s_l-1}(t_{s_l-1})\ldots\int\limits_t^{t_{s_1+3}}\psi_{s_1+2}(t_{s_1+2})\int\limits_t^{t_{s_1+2}}\psi_{s_1}(t_{s_1+1})\psi_{s_1+1}(t_{s_1+1})\times$$

$$\times \int\limits_{t}^{t_{s_1+1}} \psi_{s_1-1}(t_{s_1-1}) \ldots \int\limits_{t}^{t_2} \psi_1(t_1) d\mathbf{w}_{t_1}^{(i_1)} \ldots d\mathbf{w}_{t_{s_1-1}}^{(i_{s_1-1})} dt_{s_1+1} d\mathbf{w}_{t_{s_1+2}}^{(i_{s_1+2})} \ldots$$

$$\ldots d\mathbf{w}_{t_{s_l-1}}^{(i_{s_l-1})} dt_{s_l+1} d\mathbf{w}_{t_{s_l+2}}^{(i_{s_l+2})} \ldots d\mathbf{w}_{t_k}^{(i_k)}, \tag{7}$$

where $\psi_1(\tau), \ldots, \psi_k(\tau) \in L_2([t,T])$, the iterated integral on the right-hand side of (7) is the iterated Itô stochastic integral, $(s_l, \ldots, s_1) \in \mathrm{A}_{k,l}$,

$$\mathrm{A}_{k,l} = \left\{(s_l, \ldots, s_1): \ s_l > s_{l-1}+1, \ldots, s_2 > s_1+1; \ s_l, \ldots, s_1 = 1, \ldots, k-1\right\}, \tag{8}$$

$l = 1, 2, \ldots, [k/2]$, $i_1, \ldots, i_k = 0, 1, \ldots, m$, $[x]$ is an integer part of a real number x, $\mathbf{1}_A$ is the indicator of the set A.

For $l = 0$, we understand (7) as the following iterated Itô stochastic integral:

$$J[\psi^{(k)}]_{T,t}^{(i_1 \ldots i_k)} = \int\limits_{t}^{T} \psi_k(t_k) \ldots \int\limits_{t}^{t_2} \psi_1(t_1) d\mathbf{w}_{t_1}^{(i_1)} \ldots d\mathbf{w}_{t_k}^{(i_k)}, \tag{9}$$

where $i_1, \ldots, i_k = 0, 1, \ldots, m$ and $\mathbf{w}_{\tau}^{(0)} = \tau$.

Let us denote

$$J[\psi^{(k)}]_{T,t}^{(i_1 \ldots i_k)} + \sum_{r=1}^{[k/2]} \frac{1}{2^r} \sum_{(s_r, \ldots, s_1) \in \mathrm{A}_{k,r}} J[\psi^{(k)}]_{T,t}^{(i_1 \ldots i_k)[s_r, \ldots, s_1]} \stackrel{\mathsf{def}}{=} \bar{J}^*[\psi^{(k)}]_{T,t}^{(i_1 \ldots i_k)}, \tag{10}$$

where $\psi_1(\tau), \ldots, \psi_k(\tau) \in L_2([t,T])$, $J[\psi^{(k)}]_{T,t}^{(i_1 \ldots i_k)}$ is the iterated Itô stochastic integral (9), $\sum\limits_{\emptyset}$ is supposed to be equal to zero.

Theorem 4 [16] (Sect. 2.22), [31], [32]. *Assume that the CONS* $\{\phi_j(x)\}_{j=0}^{\infty}$ *in* $L_2([t,T])$ *and* $\psi_1(\tau), \ldots, \psi_k(\tau) \in L_2([t,T])$ *are such that*

$$\lim_{p \to \infty} \sum_{j_{q_1}, \ldots, j_{q_{k-2r}}=0}^{p} \Bigg(\sum_{j_{g_1}, j_{g_3}, \ldots, j_{g_{2r-1}}=0}^{p} C_{j_k \ldots j_1} \Bigg|_{j_{g_1}=j_{g_2}, \ldots, j_{g_{2r-1}}=j_{g_{2r}}} -$$

$$-\frac{1}{2^r} \prod_{l=1}^{r} \mathbf{1}_{\{g_{2l}=g_{2l-1}+1\}} C_{j_k \ldots j_1} \Bigg|_{(j_{g_2} j_{g_1}) \curvearrowright (\cdot) \ldots (j_{g_{2r}} j_{g_{2r-1}}) \curvearrowright (\cdot), j_{g_1} = \ j_{g_2}, \ldots, j_{g_{2r-1}} = \ j_{g_{2r}}} \Bigg)^2 = 0 \tag{11}$$

for all $r = 1, 2, \ldots, [k/2]$ *and for all possible* $g_1, g_2, \ldots, g_{2r-1}, g_{2r}$ *(see* (5)*). Then, for the sum* $\bar{J}^*[\psi^{(k)}]_{T,t}^{(i_1\ldots i_k)}$ *of iterated Itô stochastic integrals defined by* (10) *the following expansion:*

$$\bar{J}^*[\psi^{(k)}]_{T,t}^{(i_1\ldots i_k)} = \underset{p\to\infty}{\text{l.i.m.}} \sum_{j_1,\ldots,j_k=0}^{p} C_{j_k\ldots j_1} \prod_{l=1}^{k} \zeta_{j_l}^{(i_l)}$$

that converges in the mean-square sense is valid, where $C_{j_k\ldots j_1}$ *has the form* (2), l.i.m. *is a limit in the mean-square sense,* $i_1, \ldots, i_k = 0, 1, \ldots, m$,

$$\zeta_j^{(i)} = \int_t^T \phi_j(\tau) d\mathbf{w}_\tau^{(i)}$$

are independent standard Gaussian random variables for various i *or* j *(in the case* $i \neq 0$*), and* $\mathbf{w}_\tau^{(0)} = \tau$.

Let us formulate the statements on connection between iterated Stratonovich and Itô stochastic integrals (1) and (9) of arbitrary multiplicity k ($k \in \mathbf{N}$).

Theorem 5 [38] (1997) (also see [16] (Sect. 2.4.1, Theorem 2.12)). *Suppose that* $\psi_1(\tau), \ldots, \psi_k(\tau)$ *are continuous functions on* $[t, T]$. *Then*

$$J^*[\psi^{(k)}]_{T,t}^{(i_1\ldots i_k)} = J[\psi^{(k)}]_{T,t}^{(i_1\ldots i_k)} + \sum_{r=1}^{[k/2]} \frac{1}{2^r} \sum_{(s_r,\ldots,s_1)\in \mathrm{A}_{k,r}} J[\psi^{(k)}]_{T,t}^{(i_1\ldots i_k)[s_r,\ldots,s_1]} \quad \text{w. p. 1}, \tag{12}$$

where $i_1, \ldots, i_k = 0, 1, \ldots, m$, $\sum\limits_{\emptyset}$ *is supposed to be equal to zero, w. p.* 1 *(here and further) means with probability* 1.

Theorem 6 [16] (Sect. 4.6, Proposition 4.3). *Suppose that* $\psi_1(\tau), \ldots, \psi_k(\tau)$ *are continuous functions on* $[t, T]$. *Then*

$$J[\psi^{(k)}]_{T,t}^{(i_1\ldots i_k)} = J^*[\psi^{(k)}]_{T,t}^{(i_1\ldots i_k)} + \sum_{r=1}^{[k/2]} \frac{(-1)^r}{2^r} \sum_{(s_r,\ldots,s_1)\in \mathrm{A}_{k,r}} J^*[\psi^{(k)}]_{T,t}^{(i_1\ldots i_k)[s_r,\ldots,s_1]} \quad \text{w. p. 1}, \tag{13}$$

where $J^*[\psi^{(k)}]_{T,t}^{(i_1\ldots i_k)[s_r,\ldots,s_1]}$ *is defined similarly to* $J[\psi^{(k)}]_{T,t}^{(i_1\ldots i_k)[s_r,\ldots,s_1]}$, *only on the right-hand side of* (7) *all Itô integrals are replaced by Stratonovich integrals; another notations as in Theorem* 5.

Note that the condition of continuity of the functions $\psi_1(\tau),\ldots,\psi_k(\tau)$ is related to the definition of the Stratonovich stochastic integral that we use [35] (also see [16], Sect. 2.1.1).

Combining Theorems 4 and 5, we obtain the following result.

Theorem 7 [16] (Sect. 2.22), [31], [32]. *Suppose that the CONS* $\{\phi_j(x)\}_{j=0}^{\infty}$ *in* $L_2([t,T])$ *and continuous functions* $\psi_1(\tau),\ldots,\psi_k(\tau)$ *on* $[t,T]$ *are such that the condition* (11) *is satisfied. Then*

$$J^*[\psi^{(k)}]_{T,t}^{(i_1\ldots i_k)}=\underset{p\to\infty}{\text{l.i.m.}}\sum_{j_1,\ldots,j_k=0}^{p}C_{j_k\ldots j_1}\prod_{l=1}^{k}\zeta_{j_l}^{(i_l)}, \tag{14}$$

where $J^*[\psi^{(k)}]_{T,t}^{(i_1\ldots i_k)}$ *is the iterated Stratonovich stochastic integral* (1); *another notations as in Theorem* 4.

Suppose that $k>2r$. Consider the following variant of sufficient condition under which the condition (11) is satisfied (see [16], Sect. 2.29, 2.30):

$$\exists\quad \lim_{p,q\to\infty}\sum_{j_{q_1},\ldots,j_{q_{k-2r}}=0}^{q}\Biggl(\sum_{j_{g_1},j_{g_3},\ldots,j_{g_{2r-1}}=0}^{p}C_{j_k\ldots j_1}\Biggl|_{j_{g_1}=j_{g_2},\ldots,j_{g_{2r-1}}=j_{g_{2r}}}-$$

$$-\frac{1}{2^r}\prod_{l=1}^{r}\mathbf{1}_{\{g_{2l}=g_{2l-1}+1\}}C_{j_k\ldots j_1}\Biggl|_{(j_{g_2}j_{g_1})\curvearrowright(\cdot)\ldots(j_{g_{2r}}j_{g_{2r-1}})\curvearrowright(\cdot),j_{g_1}=\ j_{g_2},\ldots,j_{g_{2r-1}}=\ j_{g_{2r}}}\Biggr)^2<\infty \tag{15}$$

for all $r=1,2,\ldots,[k/2]$, where notations are the same as in (11). Recall that the case $k=2r$ of (11) is proved in [39]-[41] (also see [14], Sect. 3.2 or [16], Sect. 2.27.4).

Theorem 8 [16] (Sect. 1.11). *Suppose that* $\psi_1(\tau),\ldots,\psi_k(\tau)\in L_2([t,T])$ *and* $\{\phi_j(x)\}_{j=0}^{\infty}$ *is an arbitrary CONS in* $L_2([t,T])$. *Then*

$$J[\psi^{(k)}]_{T,t}^{(i_1\ldots i_k)}=\underset{p_1,\ldots,p_k\to\infty}{\text{l.i.m.}}\sum_{j_1=0}^{p_1}\ldots\sum_{j_k=0}^{p_k}C_{j_k\ldots j_1}J'[\phi_{j_1}\ldots\phi_{j_k}]_{T,t}^{(i_1\ldots i_k)}, \tag{16}$$

where $i_1,\ldots,i_k=0,1,\ldots,m$, $J[\psi^{(k)}]_{T,t}^{(i_1\ldots i_k)}$ *is the iterated Itô stochastic integral* (9), $J'[\phi_{j_1}\ldots\phi_{j_k}]_{T,t}^{(i_1\ldots i_k)}$ *is the multiple Wiener stochastic integral defined as in* [1] *but for the case of a multidimensional Wiener process* (*see* [16], *Sect.* 1.11 *for details*), $C_{j_k\ldots j_1}$ *is the Fourier coefficient* (2); *another notations as in Theorem* 1.

3 Main Results

3.1 The Connection of Condition (11) with the Concept of Limiting Traces from the Work of G.W. Johnson and G. Kallianpur [5]

Assume that $\{\phi_j(x)\}_{j=0}^{\infty}$ is an arbitrary CONS in $L_2([t,T])$ and $\psi_1(\tau),\ldots,\psi_k(\tau)$ $\in L_2([t,T])$.

By analogy, with Definition 2.5 in [4], we define the rth limiting trace of the function $K(t_1,\ldots,t_k)\in L_2([t,T]^k)$ of type (3) by the following expression:

$$T_{g_1,\ldots,g_{2r}}^{k-2r}K(t_{q_1},\ldots,t_{q_{k-2r}})\stackrel{\sf def}{=}\lim_{p\to\infty}T_{g_1,\ldots,g_{2r}}^{k-2r,p}K(t_{q_1},\ldots,t_{q_{k-2r}}) \tag{17}$$

in $L_2([t,T]^{k-2r})$ (here and further $L_2([t,T]^0)\stackrel{\sf def}{=}\mathbf{R}$), where

$$T_{g_1,\ldots,g_{2r}}^{k-2r,p}K(t_{q_1},\ldots,t_{q_{k-2r}})=$$

$$=\sum_{j_{q_1},\ldots,j_{q_{k-2r}}=0}^{p}\ \sum_{j_{g_1},j_{g_3},\ldots,j_{g_{2r-1}}=0}^{p}C_{j_k\ldots j_1}\Bigl|_{j_{g_1}=j_{g_2},\ldots,j_{g_{2r-1}}=j_{g_{2r}}}\cdot\phi_{j_{q_1}}(t_{q_1})\ldots\phi_{j_{q_{k-2r}}}(t_{q_{k-2r}}),$$

where $r=1,2,\ldots,[k/2]$ and $\{g_1,g_2,\ldots,g_{2r-1},g_{2r},q_1,\ldots,q_{k-2r}\}=\{1,2,\ldots,k\}$ (see (5)), $C_{j_k\ldots j_1}$ is the Fourier coefficient (2). In addition, we write

$$T_{g_1,\ldots,g_{2r}}^{k-2r}K(t_{q_1},\ldots,t_{q_{k-2r}})=K(t_1,\ldots,t_k)$$

for $r=0$.

Note that in [5] the Wiener process is scalar, while in this article the Wiener process is a multidimensional process with independent components. One of the main results of work [5] (Theorem 5.1) is obtained under the condition of existence of limiting traces (see Definition 2.5 in [4]).

Further, we will show that the condition (11) is a necessary and sufficient condition for the existence of limiting traces (17) (for all $r=1,2,\ldots,[k/2]$ and for all possible $g_1,g_2,\ldots,g_{2r-1},g_{2r}$ (see (5))) for the case of a multidimensional Wiener process.

Here it is also appropriate to recall the formula (see the proof in [16], Sect. 2.18)

$$\underset{p_1,\ldots,p_k\to\infty}{\text{l.i.m.}}\sum_{j_1=0}^{p_1}\ldots\sum_{j_k=0}^{p_k}C_{j_k\ldots j_1}\zeta_{j_1}^{(i_1)}\ldots\zeta_{j_k}^{(i_k)}=J[\psi^{(k)}]_{T,t}^{(i_1\ldots i_k)}+$$

$$+\sum_{r=1}^{[k/2]}\sum_{\substack{(\{\{g_1,g_2\},\ldots,\{g_{2r-1},g_{2r}\}\},\{q_1,\ldots,q_{k-2r}\})\\ \{g_1,g_2,\ldots,g_{2r-1},g_{2r},q_1,\ldots,q_{k-2r}\}=\{1,2,\ldots,k\}}}\prod_{s=1}^{r}\mathbf{1}_{\{i_{g_{2s-1}}=\ i_{g_{2s}}\neq 0\}}\times$$

$$\times\underset{p_1,\ldots,p_k\to\infty}{\text{l.i.m.}}\sum_{j_1=0}^{p_1}\ldots\sum_{j_k=0}^{p_k}C_{j_k\ldots j_1}\prod_{s=1}^{r}\mathbf{1}_{\{j_{g_{2s-1}}=\ j_{g_{2s}}\}}J'[\phi_{j_{q_1}}\ldots\phi_{j_{q_{k-2r}}}]_{T,t}^{(i_{q_1}\ldots i_{q_{k-2r}})}\qquad(18)$$

w. p. 1, where $J'[\phi_{j_{q_1}}\ldots\phi_{j_{q_{k-2r}}}]_{T,t}^{(i_{q_1}\ldots i_{q_{k-2r}})}$ is the multiple Wiener stochastic integral defined as in [1] but for the case of a multidimensional Wiener process (see [16], Sect. 1.11 for details), $J'[\phi_{j_{q_1}}\ldots\phi_{j_{q_{k-2r}}}]_{T,t}^{(i_{q_1}\ldots i_{q_{k-2r}})}\stackrel{\mathsf{def}}{=}1$ for $k=2r$, $J[\psi^{(k)}]_{T,t}^{(i_1\ldots i_k)}$ is the iterated Itô stochastic integral (9), $C_{j_k\ldots j_1}$ is the Fourier coefficient (2), $\{\phi_j(x)\}_{j=0}^{\infty}$ is an arbitrary CONS in $L_2([t,T])$, and $\psi_1(\tau),\ldots,\psi_k(\tau)\in L_2([t,T])$.

The equality (18) is an analogue of the formula (5.1) (see [5], Theorem 5.1) for the case of a multidimensional Wiener process.

We have

$$T_{g_1,\ldots,g_{2r}}^{k-2r,p}K(t_{q_1},\ldots,t_{q_{k-2r}})=$$

$$=\sum_{j_{q_1},\ldots,j_{q_{k-2r}}=0}^{p}\Biggl(\sum_{j_{g_1},j_{g_3},\ldots,j_{g_{2r-1}}=0}^{p}C_{j_k\ldots j_1}\Biggr|_{j_{g_1}=j_{g_2},\ldots,j_{g_{2r-1}}=j_{g_{2r}}}-$$

$$-\frac{1}{2^r}\prod_{l=1}^{r}\mathbf{1}_{\{g_{2l}=g_{2l-1}+1\}}C_{j_k\ldots j_1}\Biggr|_{(j_{g_2}j_{g_1})\curvearrowright(\cdot)\ldots(j_{g_{2r}}j_{g_{2r-1}})\curvearrowright(\cdot),j_{g_1}=\ j_{g_2},\ldots,j_{g_{2r-1}}=\ j_{g_{2r}}}\Biggr)\times$$

$$\times\phi_{j_{q_1}}(t_{q_1})\ldots\phi_{j_{q_{k-2r}}}(t_{q_{k-2r}})+$$

$$+\frac{1}{2^r}\prod_{l=1}^{r}\mathbf{1}_{\{g_{2l}=g_{2l-1}+1\}}\sum_{j_{q_1},\ldots,j_{q_{k-2r}}=0}^{p}C_{j_k\ldots j_1}\Biggr|_{(j_{g_2}j_{g_1})\curvearrowright(\cdot)\ldots(j_{g_{2r}}j_{g_{2r-1}})\curvearrowright(\cdot),j_{g_1}=\ j_{g_2},\ldots,j_{g_{2r-1}}=\ j_{g_{2r}}}\times$$

$$\times\phi_{j_{q_1}}(t_{q_1})\ldots\phi_{j_{q_{k-2r}}}(t_{q_{k-2r}})\stackrel{\mathsf{def}}{=}$$

$$\stackrel{\sf def}{=} F^{(p)}_{g_1,\ldots,g_{2r}}(t_{q_1},\ldots,t_{q_{k-2r}}) + G^{(p)}_{g_1,\ldots,g_{2r}}(t_{q_1},\ldots,t_{q_{k-2r}}). \tag{19}$$

Denote

$$C_{j_k\ldots j_1}\Bigl|_{(j_{g_2}j_{g_1})\curvearrowright(\cdot)\ldots(j_{g_{2r}}j_{g_{2r-1}})\curvearrowright(\cdot),j_{g_1}=\,j_{g_2},\ldots,j_{g_{2r-1}}=\,j_{g_{2r}}} \stackrel{\sf def}{=} C^{g_1\ldots g_{2r}}_{j_{q_{k-2r}}\ldots j_{q_1}}.$$

Using Fubini's Theorem, we obtain (see the derivation of (135) in [14])

$$\int\limits_t^T h_k(t_k)\ldots\int\limits_t^{t_{l+2}} h_{l+1}(t_{l+1})\int\limits_t^{t_{l+1}} h_l(t_l)\int\limits_t^{t_l} h_{l-1}(t_{l-1})\ldots\int\limits_t^{t_2} h_1(t_1)dt_1\ldots$$

$$\ldots dt_{l-1}dt_l dt_{l+1}\ldots dt_k =$$

$$=\int\limits_t^T h_k(t_k)\ldots\int\limits_t^{t_{l+2}} h_{l+1}(t_{l+1})\left(\int\limits_t^{t_{l+1}} h_l(t_l)dt_l\right)\int\limits_t^{t_{l+1}} h_{l-1}(t_{l-1})\ldots$$

$$\ldots\int\limits_t^{t_2} h_1(t_1)dt_1\ldots dt_{l-1}dt_{l+1}\ldots dt_k-$$

$$-\int\limits_t^T h_k(t_k)\ldots\int\limits_t^{t_{l+2}} h_{l+1}(t_{l+1})\int\limits_t^{t_{l+1}} h_{l-1}(t_{l-1})\left(\int\limits_t^{t_{l-1}} h_l(t_l)dt_l\right)\int\limits_t^{t_{l-1}} h_{l-2}(t_{l-2})\ldots$$

$$\ldots\int\limits_t^{t_2} h_1(t_1)dt_1\ldots dt_{l-2}dt_{l-1}dt_{l+1}\ldots dt_k, \tag{20}$$

where $2<l<k-1$ and $h_1(\tau),\ldots,h_k(\tau)\in L_2([t,T])$.

By analogy, with (20) we have for $l=k$ (see the derivation of (136) in [14])

$$\int\limits_t^T h_l(t_l)\int\limits_t^{t_l} h_{l-1}(t_{l-1})\ldots\int\limits_t^{t_2} h_1(t_1)dt_1\ldots dt_{l-1}dt_l =$$

$$=\left(\int\limits_t^T h_l(t_l)dt_l\right)\int\limits_t^T h_{l-1}(t_{l-1})\ldots\int\limits_t^{t_2} h_1(t_1)dt_1\ldots dt_{l-1}-$$

$$-\int\limits_{t}^{T} h_{l-1}(t_{l-1})\left(\int\limits_{t}^{t_{l-1}} h_l(t_l)dt_l\right)\int\limits_{t}^{t_{l-1}} h_{l-2}(t_{l-2})\ldots\int\limits_{t}^{t_2} h_1(t_1)dt_1\ldots dt_{l-1}. \qquad (21)$$

We will assume that for $l=1$ the transformation (20) is not carried out since

$$\int\limits_{t}^{t_2} h_1(t_1)dt_1$$

is the innermost integral on the left-hand side of (20).

Applying transformations (20) and (21) iteratively to $C^{g_1\ldots g_{2r}}_{j_{q_{k-2r}}\ldots j_{q_1}}$ for integrations not involving the basis functions $\phi_{j_{q_1}},\ldots,\phi_{j_{q_{k-2r}}}$, we obtain

$$C^{g_1\ldots g_{2r}}_{j_{q_{k-2r}}\ldots j_{q_1}}=\sum_{d=1}^{2^r}(-1)^{d-1}\left(\hat{C}^{(d)g_1\ldots g_{2r}}_{j_{q_{k-2r}}\ldots j_{q_1}}-\bar{C}^{(d)g_1\ldots g_{2r}}_{j_{q_{k-2r}}\ldots j_{q_1}}\right), \qquad (22)$$

where some terms in the sum

$$\sum_{d=1}^{2^r}$$

can be identically equal to zero due to the remark to (20), (21).

Using (22), we get

$$G^{(p)}_{g_1,\ldots,g_{2r}}(t_{q_1},\ldots,t_{q_{k-2r}})=\frac{1}{2^r}\prod_{l=1}^{r}\mathbf{1}_{\{g_{2l}=g_{2l-1}+1\}}\times$$

$$\times\sum_{d=1}^{2^r}(-1)^{d-1}\Bigg(\sum_{j_{q_1},\ldots,j_{q_{k-2r}}=0}^{p}\hat{C}^{(d)g_1\ldots g_{2r}}_{j_{q_{k-2r}}\ldots j_{q_1}}\cdot\phi_{j_{q_1}}(t_{q_1})\ldots\phi_{j_{q_{k-2r}}}(t_{q_{k-2r}})-$$

$$-\sum_{j_{q_1},\ldots,j_{q_{k-2r}}=0}^{p}\bar{C}^{(d)g_1\ldots g_{2r}}_{j_{q_{k-2r}}\ldots j_{q_1}}\cdot\phi_{j_{q_1}}(t_{q_1})\ldots\phi_{j_{q_{k-2r}}}(t_{q_{k-2r}})\Bigg)\ \to$$

$$\to\frac{1}{2^r}\prod_{l=1}^{r}\mathbf{1}_{\{g_{2l}=g_{2l-1}+1\}}\times$$

$$\times\sum_{d=1}^{2^r}(-1)^{d-1}\left(\hat{F}^{(d)}_{g_1,\ldots,g_{2r}}(t_{q_1},\ldots,t_{q_{k-2r}})-\bar{F}^{(d)}_{g_1,\ldots,g_{2r}}(t_{q_1},\ldots,t_{q_{k-2r}})\right)\stackrel{\text{def}}{=}$$

$$\stackrel{\mathsf{def}}{=} G_{g_1,\ldots,g_{2r}}(t_{q_1},\ldots,t_{q_{k-2r}}) \ \text{ if } \ p\to\infty \ \ (\text{in } L_2([t,T]^{k-2r})), \tag{23}$$

where $G_{g_1,\ldots,g_{2r}}(t_{q_1},\ldots,t_{q_{k-2r}})$, $\hat{F}^{(d)}_{g_1,\ldots,g_{2r}}(t_{q_1},\ldots,t_{q_{k-2r}})$, $\bar{F}^{(d)}_{g_1,\ldots,g_{2r}}(t_{q_1},\ldots,t_{q_{k-2r}}) \in L_2([t,T]^{k-2r})$, $d=1,\ldots,2^r$.

Futhermore,

$$\left\|F^{(p)}_{g_1,\ldots,g_{2r}}\right\|^2_{L_2([t,T]^{k-2r})} =$$

$$= \int\limits_{[t,T]^{k-2r}} \Biggl(\sum_{j_{q_1},\ldots,j_{q_{k-2r}}=0}^{p} \Biggl(\sum_{j_{g_1},j_{g_3},\ldots,j_{g_{2r-1}}=0}^{p} C_{j_k\ldots j_1}\Bigg|_{j_{g_1}=j_{g_2},\ldots,j_{g_{2r-1}}=j_{g_{2r}}} -$$

$$-\frac{1}{2^r}\prod_{l=1}^{r}\mathbf{1}_{\{g_{2l}=g_{2l-1}+1\}}C_{j_k\ldots j_1}\Bigg|_{(j_{g_2}j_{g_1})\curvearrowright(\cdot)\ldots(j_{g_{2r}}j_{g_{2r-1}})\curvearrowright(\cdot),j_{g_1}=\ j_{g_2},\ldots,j_{g_{2r-1}}=\ j_{g_{2r}}}\Biggr)\times$$

$$\times\phi_{j_{q_1}}(t_{q_1})\ldots\phi_{j_{q_{k-2r}}}(t_{q_{k-2r}})\Biggr)^2 dt_{q_1}\ldots dt_{q_{k-2r}} =$$

$$= \sum_{j_{q_1},\ldots,j_{q_{k-2r}}=0}^{p}\Biggl(\sum_{j_{g_1},j_{g_3},\ldots,j_{g_{2r-1}}=0}^{p} C_{j_k\ldots j_1}\Bigg|_{j_{g_1}=j_{g_2},\ldots,j_{g_{2r-1}}=j_{g_{2r}}} -$$

$$-\frac{1}{2^r}\prod_{l=1}^{r}\mathbf{1}_{\{g_{2l}=g_{2l-1}+1\}}C_{j_k\ldots j_1}\Bigg|_{(j_{g_2}j_{g_1})\curvearrowright(\cdot)\ldots(j_{g_{2r}}j_{g_{2r-1}})\curvearrowright(\cdot),j_{g_1}=\ j_{g_2},\ldots,j_{g_{2r-1}}=\ j_{g_{2r}}}\Biggr)^2.$$

This means that the condition (11) is equivalent to

$$\lim_{p\to\infty}\left\|F^{(p)}_{g_1,\ldots,g_{2r}}\right\|^2_{L_2([t,T]^{k-2r})} = 0. \tag{24}$$

Suppose that the condition (11) (or (24)) is fulfilled. Applying (19), (23), and (24), we obtain

$$\left\|T^{k-2r,p}_{g_1,\ldots,g_{2r}}K - G_{g_1,\ldots,g_{2r}}\right\|_{L_2([t,T]^{k-2r})} =$$

$$= \left\|F^{(p)}_{g_1,\ldots,g_{2r}} + G^{(p)}_{g_1,\ldots,g_{2r}} - G_{g_1,\ldots,g_{2r}}\right\|_{L_2([t,T]^{k-2r})} \le$$

$$\leq \left\| F^{(p)}_{g_1,\ldots,g_{2r}} \right\|_{L_2([t,T]^{k-2r})} + \left\| G^{(p)}_{g_1,\ldots,g_{2r}} - G_{g_1,\ldots,g_{2r}} \right\|_{L_2([t,T]^{k-2r})} \to 0$$

if $p \to \infty$. Thus, the limiting trace $T^{k-2r}_{g_1,\ldots,g_{2r}} K(t_{q_1},\ldots,t_{q_{k-2r}})$ exists under the condition (11), i.e.

$$T^{k-2r}_{g_1,\ldots,g_{2r}} K(t_{q_1},\ldots,t_{q_{k-2r}}) = \lim_{p\to\infty} T^{k-2r,p}_{g_1,\ldots,g_{2r}} K(t_{q_1},\ldots,t_{q_{k-2r}}) =$$

$$= G_{g_1,\ldots,g_{2r}}(t_{q_1},\ldots,t_{q_{k-2r}}) \tag{25}$$

in $L_2([t,T]^{k-2r})$ for all $r = 1, 2, \ldots, [k/2]$ and all possible $g_1, g_2, \ldots, g_{2r-1}, g_{2r}$ (see (5)), where $G_{g_1,\ldots,g_{2r}}(t_{q_1},\ldots,t_{q_{k-2r}}) \in L_2([t,T]^{k-2r})$.

Further, we assume that the condition (25) is fulfilled. Using (25) and (19), (23), we obtain

$$\left\| F^{(p)}_{g_1,\ldots,g_{2r}} \right\|_{L_2([t,T]^{k-2r})} = \left\| T^{k-2r,p}_{g_1,\ldots,g_{2r}} K - G^{(p)}_{g_1,\ldots,g_{2r}} \right\|_{L_2([t,T]^{k-2r})} =$$

$$= \left\| T^{k-2r,p}_{g_1,\ldots,g_{2r}} K - G_{g_1,\ldots,g_{2r}} + G_{g_1,\ldots,g_{2r}} - G^{(p)}_{g_1,\ldots,g_{2r}} \right\|_{L_2([t,T]^{k-2r})} \leq$$

$$\leq \left\| T^{k-2r,p}_{g_1,\ldots,g_{2r}} K - G_{g_1,\ldots,g_{2r}} \right\|_{L_2([t,T]^{k-2r})} + \left\| G_{g_1,\ldots,g_{2r}} - G^{(p)}_{g_1,\ldots,g_{2r}} \right\|_{L_2([t,T]^{k-2r})} \to 0$$

if $p \to \infty$. Thus, the condition (24) is satisfied under the condition (25).

3.2 New Representations of the Hu–Meyer Formulas for the Case of a Multidimensional Wiener Process. Connection with Theorems 4 and 7

Suppose that $\{\phi_j(x)\}_{j=0}^{\infty}$ is an arbitrary CONS in $L_2([t,T])$ and $\Phi(t_1,\ldots,t_k) \in L_2([t,T]^k)$. Let us generalize the definition of the limiting trace from the previous section.

We define the rth limiting trace of the function $\Phi(t_1,\ldots,t_k) \in L_2([t,T]^k)$ by the following expression:

$$T^{k-2r}_{g_1,\ldots,g_{2r}} \Phi(t_{q_1},\ldots,t_{q_{k-2r}}) \stackrel{\mathsf{def}}{=} \lim_{p\to\infty} T^{k-2r,p}_{g_1,\ldots,g_{2r}} \Phi(t_{q_1},\ldots,t_{q_{k-2r}}) \tag{26}$$

in $L_2([t,T]^{k-2r})$ (here and further $L_2([t,T]^0) \stackrel{\mathsf{def}}{=} \mathbf{R}$), where

$$T_{g_1,\ldots,g_{2r}}^{k-2r,p}\Phi(t_{q_1},\ldots,t_{q_{k-2r}})=$$

$$=\sum_{j_{q_1},\ldots,j_{q_{k-2r}}=0}^{p}\ \sum_{j_{g_1},j_{g_3},\ldots,j_{g_{2r-1}}=0}^{p}C_{j_k\ldots j_1}\Bigl|_{j_{g_1}=j_{g_2},\ldots,j_{g_{2r-1}}=j_{g_{2r}}}\cdot\phi_{j_{q_1}}(t_{q_1})\ldots\phi_{j_{q_{k-2r}}}(t_{q_{k-2r}}),$$

where $r=1,2,\ldots,[k/2]$ and $\{g_1,g_2,\ldots,g_{2r-1},g_{2r},q_1,\ldots,q_{k-2r}\}=\{1,2,\ldots,k\}$ (see (5)),

$$C_{j_k\ldots j_1}=\int\limits_{[t,T]^k}\Phi(t_1,\ldots,t_k)\prod_{l=1}^{k}\phi_{j_l}(t_l)dt_1\ldots dt_k \tag{27}$$

is the Fourier coefficient. Also we write $T_{g_1,\ldots,g_{2r}}^{k-2r}\Phi(t_{q_1},\ldots,t_{q_{k-2r}})=\Phi(t_1,\ldots,t_k)$ and $T_{g_1,\ldots,g_{2r}}^{k-2r,p}\Phi(t_{q_1},\ldots,t_{q_{k-2r}})=\Phi_p(t_1,\ldots,t_k)$ for $r=0$, where

$$\Phi_p(t_1,\ldots,t_k)=\sum_{j_1,\ldots,j_k=0}^{p}C_{j_k\ldots j_1}\phi_{j_1}(t_1)\ldots\phi_{j_k}(t_k).$$

Let us consider a prelimit variant of the formula (18) for the case $p_1=\ldots=p_k=p$ (see the proof in [16], Sect. 2.18) and replace $K(t_1,\ldots,t_k)$ with $\Phi(t_1,\ldots,t_k)$ in it. Thus, we have

$$\sum_{j_1,\ldots,j_k=0}^{p}C_{j_k\ldots j_1}\prod_{l=1}^{k}\zeta_{j_l}^{(i_l)}=\sum_{j_1,\ldots,j_k=0}^{p}C_{j_k\ldots j_1}J'[\phi_{j_1}\ldots\phi_{j_k}]_{T,t}^{(i_1\ldots i_k)}+$$

$$+\sum_{r=1}^{[k/2]}\ \sum_{\substack{(\{\{g_1,g_2\},\ldots,\{g_{2r-1},g_{2r}\}\},\{q_1,\ldots,q_{k-2r}\})\\ \{g_1,g_2,\ldots,g_{2r-1},g_{2r},q_1,\ldots,q_{k-2r}\}=\{1,2,\ldots,k\}}}\ \prod_{s=1}^{r}\mathbf{1}_{\{i_{g_{2s-1}}=\ i_{g_{2s}}\neq 0\}}\times$$

$$\times\sum_{j_1,\ldots,j_k=0}^{p}C_{j_k\ldots j_1}\mathbf{1}_{\{j_{g_{2s-1}}=\ j_{g_{2s}}\}}J'[\phi_{j_{q_1}}\ldots\phi_{j_{q_{k-2r}}}]_{T,t}^{(i_{q_1}\ldots i_{q_{k-2r}})}\quad\text{w. p. 1,} \tag{28}$$

where $k\geq 2$, $J'[\phi_{j_1}\ldots\phi_{j_k}]_{T,t}^{(i_1\ldots i_k)}$, $J'[\phi_{j_{q_1}}\ldots\phi_{j_{q_{k-2r}}}]_{T,t}^{(i_{q_1}\ldots i_{q_{k-2r}})}$ are multiple Wiener stochastic integrals defined as in [1] but for the case of a multidimensional Wiener process (see [16], Sect. 1.11 for details), $J'[\phi_{j_{q_1}}\ldots\phi_{j_{q_{k-2r}}}]_{T,t}^{(i_{q_1}\ldots i_{q_{k-2r}})}\stackrel{\mathsf{def}}{=}1$ for $k=2r$, and $C_{j_k\ldots j_1}$ is defined by (27).

Note that sometimes the multiple Stratonovich stochastic integral for $\Phi(t_1,\ldots,t_k)\in L_2([t,T]^k)$ is defined as the limit in probability as $p\to\infty$ of the following expression (the case of a scalar standard Wiener process):

$$\sum_{j_1,\ldots,j_k=0}^{p} C_{j_k\ldots j_1}\prod_{l=1}^{k}\zeta_{j_l}^{(i)},$$

where $i_1=\ldots=i_k=i$ (see, for example, Definition 5.9 in [2]).

By analogy, with [2], we define the multiple Stratonovich stochastic integral for $\Phi(t_1,\ldots,t_k)\in L_2([t,T]^k)$ (the case of a multidimensional Wiener process) as the following mean-square limit:

$$\hat{J}^S[\Phi]_{T,t}^{(i_1\ldots i_k)}=\underset{p\to\infty}{\text{l.i.m.}}\ \hat{J}_p^S[\Phi]_{T,t}^{(i_1\ldots i_k)}, \tag{29}$$

where

$$\hat{J}_p^S[\Phi]_{T,t}^{(i_1\ldots i_k)}=\sum_{j_1,\ldots,j_k=0}^{p} C_{j_k\ldots j_1}\prod_{l=1}^{k}\zeta_{j_l}^{(i_l)}, \tag{30}$$

$C_{j_k\ldots j_1}$ is the Fourier coefficient defined by (27).

Let us rewrite (28) in the form

$$\hat{J}_p^S[\Phi]_{T,t}^{(i_1\ldots i_k)}=J'[\Phi_p]_{T,t}^{(i_1\ldots i_k)}+$$

$$+\sum_{r=1}^{[k/2]}\sum_{\substack{(\{\{g_1,g_2\},\ldots,\{g_{2r-1},g_{2r}\}\},\{q_1,\ldots,q_{k-2r}\})\\ \{g_1,g_2,\ldots,g_{2r-1},g_{2r},q_1,\ldots,q_{k-2r}\}=\{1,2,\ldots,k\}}}\prod_{s=1}^{r}\mathbf{1}_{\{i_{g_{2s-1}}=\ i_{g_{2s}}\neq 0\}}\times$$

$$\times J'\Big[T_{g_1,g_2\ldots,g_{2r-1},g_{2r}}^{k-2r,p}\Phi\Big]_{T,t}^{(i_{q_1}\ldots i_{q_{k-2r}})}\quad \text{w. p. 1}, \tag{31}$$

where $J'\Big[T_{g_1,g_2\ldots,g_{2r-1},g_{2r}}^{k-2r,p}\Phi\Big]_{T,t}^{(i_{q_1}\ldots i_{q_{k-2r}})}\stackrel{\text{def}}{=}T_{g_1,g_2\ldots,g_{2r-1},g_{2r}}^{k-2r,p}\Phi$ for $k=2r$.

Assume that all limiting traces $T_{g_1,\ldots,g_{2r}}^{k-2r}\Phi(t_{q_1},\ldots,t_{q_{k-2r}})$ (for all $r=1,2,\ldots,$ $[k/2]$ and for all possible $g_1,g_2,\ldots,g_{2r-1},g_{2r}$ (see (5))) exist.

Recall the well-known property of the multiple Wiener stochastic integral

$$\mathsf{M}\left\{\left(J'[\Phi]_{T,t}^{(i_1\ldots i_k)}\right)^2\right\}\le C_k\int\limits_{[t,T]^k}\Phi^2(t_1,\ldots,t_k)dt_1\ldots dt_k, \tag{32}$$

where $\Phi(t_1,\ldots,t_k)\in L_2([t,T]^k)$ and C_k is a constant.

Using (31), (32) and the existence of limiting traces, we have

$$\mathsf{M}\Biggl\{\Biggl(\hat{J}_p^S[\Phi]_{T,t}^{(i_1\ldots i_k)}-J'[\Phi]_{T,t}^{(i_1\ldots i_k)}-$$

$$-\sum_{r=1}^{[k/2]}\sum_{\substack{(\{\{g_1,g_2\},\ldots,\{g_{2r-1},g_{2r}\}\},\{q_1,\ldots,q_{k-2r}\})\\ \{g_1,g_2,\ldots,g_{2r-1},g_{2r},q_1,\ldots,q_{k-2r}\}=\{1,2,\ldots,k\}}}\prod_{s=1}^{r}\mathbf{1}_{\{i_{g_{2s-1}}=\ i_{g_{2s}}\neq 0\}}\times$$

$$\times J'\bigl[T_{g_1,g_2\ldots,g_{2r-1},g_{2r}}^{k-2r}\Phi\bigr]_{T,t}^{(i_{q_1}\ldots i_{q_{k-2r}})}\Biggr)^2\Biggr\}=$$

$$=\mathsf{M}\Biggl\{\Biggl(J'[\Phi_p-\Phi]_{T,t}^{(i_1\ldots i_k)}+$$

$$+\sum_{r=1}^{[k/2]}\sum_{\substack{(\{\{g_1,g_2\},\ldots,\{g_{2r-1},g_{2r}\}\},\{q_1,\ldots,q_{k-2r}\})\\ \{g_1,g_2,\ldots,g_{2r-1},g_{2r},q_1,\ldots,q_{k-2r}\}=\{1,2,\ldots,k\}}}\prod_{s=1}^{r}\mathbf{1}_{\{i_{g_{2s-1}}=\ i_{g_{2s}}\neq 0\}}\times$$

$$\times J'\bigl[T_{g_1,g_2\ldots,g_{2r-1},g_{2r}}^{k-2r,p}\Phi-T_{g_1,g_2\ldots,g_{2r-1},g_{2r}}^{k-2r}\Phi\bigr]_{T,t}^{(i_{q_1}\ldots i_{q_{k-2r}})}\Biggr)^2\Biggr\}\leq$$

$$\leq C_k'\Biggl(\bigl\|\Phi_p-\Phi\bigr\|_{L_2([t,T]^k)}^2+$$

$$+\sum_{r=1}^{[k/2]}\sum_{\substack{(\{\{g_1,g_2\},\ldots,\{g_{2r-1},g_{2r}\}\},\{q_1,\ldots,q_{k-2r}\})\\ \{g_1,g_2,\ldots,g_{2r-1},g_{2r},q_1,\ldots,q_{k-2r}\}=\{1,2,\ldots,k\}}}\prod_{s=1}^{r}\mathbf{1}_{\{i_{g_{2s-1}}=\ i_{g_{2s}}\neq 0\}}\times$$

$$\times\bigl\|T_{g_1,g_2\ldots,g_{2r-1},g_{2r}}^{k-2r,p}\Phi-T_{g_1,g_2\ldots,g_{2r-1},g_{2r}}^{k-2r}\Phi\bigr\|_{L_2([t,T]^{k-2r})}^2\Biggr)\ \to\ 0 \tag{33}$$

if $p\to\infty$, where $J'\bigl[T_{g_1,g_2\ldots,g_{2r-1},g_{2r}}^{k-2r}\Phi\bigr]_{T,t}^{(i_{q_1}\ldots i_{q_{k-2r}})}\stackrel{\mathsf{def}}{=}T_{g_1,g_2\ldots,g_{2r-1},g_{2r}}^{k-2r}\Phi$ for $k=2r$, and C_k' is a constant.

Applying (33), we obtain the following new representation of the Hu–Meyer formula for the case of a multidimensional Wiener process:

$$\hat{J}^S[\Phi]_{T,t}^{(i_1\ldots i_k)} = J'[\Phi]_{T,t}^{(i_1\ldots i_k)} +$$

$$+\sum_{r=1}^{[k/2]} \sum_{\substack{(\{\{g_1,g_2\},\ldots,\{g_{2r-1},g_{2r}\}\},\{q_1,\ldots,q_{k-2r}\}) \\ \{g_1,g_2,\ldots,g_{2r-1},g_{2r},q_1,\ldots,q_{k-2r}\}=\{1,2,\ldots,k\}}} \prod_{s=1}^{r} \mathbf{1}_{\{i_{g_{2s-1}}=\ i_{g_{2s}}\neq 0\}} \times$$

$$\times J'\Big[T_{g_1,g_2\ldots,g_{2r-1},g_{2r}}^{k-2r}\Phi\Big]_{T,t}^{(i_{q_1}\ldots i_{q_{k-2r}})} \quad \text{w. p. 1.} \tag{34}$$

The equality (34) is consistent with Theorem 5.1 [5] (the case of a scalar Wiener process).

Further, let us obtain the inverse version of the Hu–Meyer formula (34), i.e. a formula expressing the multiple Wiener stochastic integral through the sum of multiple Stratonovich stochastic integrals.

Let us give the following definition of the limiting trace. We define the rth limiting trace of the function $\Phi(t_1,\ldots,t_k) \in L_2([t,T]^k)$ by the following expression:

$$\tilde{T}_{g_1,\ldots,g_{2r}}^{k-2r}\Phi(t_{q_1},\ldots,t_{q_{k-2r}}) \stackrel{\mathsf{def}}{=} \lim_{p\to\infty} \tilde{T}_{g_1,\ldots,g_{2r}}^{k-2r,p}\Phi(t_{q_1},\ldots,t_{q_{k-2r}})$$

in $L_2([t,T]^{k-2r})$, where

$$\tilde{T}_{g_1,\ldots,g_{2r}}^{k-2r,p}\Phi(t_{q_1},\ldots,t_{q_{k-2r}}) =$$

$$= \sum_{j_{q_1},\ldots,j_{q_{k-2r}}=0}^{p} \sum_{j_{g_1},j_{g_3},\ldots,j_{g_{2r-1}}=0}^{\infty} C_{j_k\ldots j_1}\Big|_{j_{g_1}=j_{g_2},\ldots,j_{g_{2r-1}}=j_{g_{2r}}} \cdot \phi_{j_{q_1}}(t_{q_1})\ldots\phi_{j_{q_{k-2r}}}(t_{q_{k-2r}}) =$$

$$= \sum_{j_{q_1},\ldots,j_{q_{k-2r}}=0}^{p} \tilde{C}_{j_{q_{k-2r}}\ldots j_{q_1}} \cdot \phi_{j_{q_1}}(t_{q_1})\ldots\phi_{j_{q_{k-2r}}}(t_{q_{k-2r}}), \tag{35}$$

where

$$\tilde{C}_{j_{q_{k-2r}}\ldots j_{q_1}} = \sum_{j_{g_1},j_{g_3},\ldots,j_{g_{2r-1}}=0}^{\infty} C_{j_k\ldots j_1}\Big|_{j_{g_1}=j_{g_2},\ldots,j_{g_{2r-1}}=j_{g_{2r}}} =$$

$$= \lim_{p\to\infty} \sum_{j_{g_1},j_{g_3},\ldots,j_{g_{2r-1}}=0}^{p} C_{j_k\ldots j_1}\Big|_{j_{g_1}=j_{g_2},\ldots,j_{g_{2r-1}}=j_{g_{2r}}}, \qquad (36)$$

$r = 1, 2, \ldots, [k/2]$, $\{g_1, g_2, \ldots, g_{2r-1}, g_{2r}, q_1, \ldots, q_{k-2r}\} = \{1, 2, \ldots, k\}$ (see (5)), $C_{j_k\ldots j_1}$ has the form (27). Also we write $\tilde{T}^{k-2r}_{g_1,\ldots,g_{2r}}\Phi(t_{q_1}, \ldots, t_{q_{k-2r}}) = \Phi(t_1, \ldots, t_k)$ and $\tilde{T}^{k-2r,p}_{g_1,\ldots,g_{2r}}\Phi(t_{q_1}, \ldots, t_{q_{k-2r}}) = \Phi_p(t_1, \ldots, t_k)$ for $r = 0$, where

$$\Phi_p(t_1, \ldots, t_k) = \sum_{j_1,\ldots,j_k=0}^{p} C_{j_k\ldots j_1}\phi_{j_1}(t_1)\ldots\phi_{j_k}(t_k).$$

Note that

$$\left\| T^{k-2r,p}_{g_1,g_2\ldots,g_{2r-1},g_{2r}}\Phi - \tilde{T}^{k-2r,p}_{g_1,g_2\ldots,g_{2r-1},g_{2r}}\Phi \right\|^2_{L_2([t,T]^{k-2r})} =$$

$$= \sum_{j_{q_1},\ldots,j_{q_{k-2r}}=0}^{p} \left(\sum_{j_{g_1},j_{g_3},\ldots,j_{g_{2r-1}}=0}^{p} C_{j_k\ldots j_1}\Big|_{j_{g_1}=j_{g_2},\ldots,j_{g_{2r-1}}=j_{g_{2r}}} - \tilde{C}_{j_{q_{k-2r}}\ldots j_{q_1}} \right)^2, \qquad (37)$$

and suppose that

$$\lim_{p\to\infty} \left\| T^{k-2r,p}_{g_1,g_2\ldots,g_{2r-1},g_{2r}}\Phi - \tilde{T}^{k-2r,p}_{g_1,g_2\ldots,g_{2r-1},g_{2r}}\Phi \right\|^2_{L_2([t,T]^{k-2r})} = 0 \qquad (38)$$

for all $r = 1, 2, \ldots, [k/2]$ and for all possible $g_1, g_2, \ldots, g_{2r-1}, g_{2r}$ (see (5)).

Next, we assume that all limiting traces $T^{k-2r}_{g_1,g_2\ldots,g_{2r-1},g_{2r}}\Phi$ and $\tilde{T}^{k-2r}_{g_1,g_2\ldots,g_{2r-1},g_{2r}}\Phi$ exist for all $r = 1, 2, \ldots, [k/2]$ and for all possible $g_1, g_2, \ldots, g_{2r-1}, g_{2r}$.

Applying (32) and (38), we obtain for $k > 2r$

$$\mathsf{M}\left\{ \left(J'\left[T^{k-2r}_{g_1,g_2\ldots,g_{2r-1},g_{2r}}\Phi\right]^{(i_{q_1}\ldots i_{q_{k-2r}})}_{T,t} - J'\left[\tilde{T}^{k-2r}_{g_1,g_2\ldots,g_{2r-1},g_{2r}}\Phi\right]^{(i_{q_1}\ldots i_{q_{k-2r}})}_{T,t} \right)^2 \right\} \le$$

$$\le 3\mathsf{M}\left\{ \left(J'\left[T^{k-2r}_{g_1,g_2\ldots,g_{2r-1},g_{2r}}\Phi - T^{k-2r,p}_{g_1,g_2\ldots,g_{2r-1},g_{2r}}\Phi\right]^{(i_{q_1}\ldots i_{q_{k-2r}})}_{T,t} \right)^2 \right\} +$$

$$+3\mathsf{M}\left\{\left(J'\left[T^{k-2r,p}_{g_1,g_2\ldots,g_{2r-1},g_{2r}}\Phi-\tilde{T}^{k-2r,p}_{g_1,g_2\ldots,g_{2r-1},g_{2r}}\Phi\right]^{(i_{q_1}\ldots i_{q_{k-2r}})}_{T,t}\right)^2\right\}+$$

$$+3\mathsf{M}\left\{\left(J'\left[\tilde{T}^{k-2r,p}_{g_1,g_2\ldots,g_{2r-1},g_{2r}}\Phi-\tilde{T}^{k-2r}_{g_1,g_2\ldots,g_{2r-1},g_{2r}}\Phi\right]^{(i_{q_1}\ldots i_{q_{k-2r}})}_{T,t}\right)^2\right\}\le$$

$$\le C_{k-2r}\left(\left\|T^{k-2r}_{g_1,g_2\ldots,g_{2r-1},g_{2r}}\Phi-T^{k-2r,p}_{g_1,g_2\ldots,g_{2r-1},g_{2r}}\Phi\right\|^2_{L_2([t,T]^{k-2r})}+\right.$$

$$+\left\|T^{k-2r,p}_{g_1,g_2\ldots,g_{2r-1},g_{2r}}\Phi-\tilde{T}^{k-2r,p}_{g_1,g_2\ldots,g_{2r-1},g_{2r}}\Phi\right\|^2_{L_2([t,T]^{k-2r})}+$$

$$\left.+\left\|\tilde{T}^{k-2r,p}_{g_1,g_2\ldots,g_{2r-1},g_{2r}}\Phi-\tilde{T}^{k-2r}_{g_1,g_2\ldots,g_{2r-1},g_{2r}}\Phi\right\|^2_{L_2([t,T]^{k-2r})}\right)\ \to\ 0$$

if $p\to\infty$, where C_{k-2r} is a constant. Thus, under the above conditions, we have w. p. 1

$$J'\left[T^{k-2r}_{g_1,g_2\ldots,g_{2r-1},g_{2r}}\Phi\right]^{(i_{q_1}\ldots i_{q_{k-2r}})}_{T,t}=J'\left[\tilde{T}^{k-2r}_{g_1,g_2\ldots,g_{2r-1},g_{2r}}\Phi\right]^{(i_{q_1}\ldots i_{q_{k-2r}})}_{T,t}. \tag{39}$$

Combining (34) and (39), we get

$$J'[\Phi]^{(i_1\ldots i_k)}_{T,t}=\hat{J}^S[\Phi]^{(i_1\ldots i_k)}_{T,t}-$$

$$-\sum_{r=1}^{[k/2]}\sum_{\substack{(\{\{g_1,g_2\},\ldots,\{g_{2r-1},g_{2r}\}\},\{q_1,\ldots,q_{k-2r}\})\\ \{g_1,g_2,\ldots,g_{2r-1},g_{2r},q_1,\ldots,q_{k-2r}\}=\{1,2,\ldots,k\}}}\prod_{s=1}^{r}\mathbf{1}_{\{i_{g_{2s-1}}=\ i_{g_{2s}}\neq 0\}}\times$$

$$\times J'\left[\tilde{T}^{k-2r}_{g_1,g_2\ldots,g_{2r-1},g_{2r}}\Phi\right]^{(i_{q_1}\ldots i_{q_{k-2r}})}_{T,t}\quad \text{w. p. 1.} \tag{40}$$

By iteratively applying the formula (40), we obtain the following inverse version of the Hu–Meyer formula (34):

$$J'[\Phi]_{T,t}^{(i_1\ldots i_k)} = \hat{J}^S[\Phi]_{T,t}^{(i_1\ldots i_k)} +$$

$$+\sum_{r=1}^{[k/2]}(-1)^r \sum_{\substack{(\{\{g_1,g_2\},\ldots,\{g_{2r-1},g_{2r}\}\},\{q_1,\ldots,q_{k-2r}\})\\ \{g_1,g_2,\ldots,g_{2r-1},g_{2r},q_1,\ldots,q_{k-2r}\}=\{1,2,\ldots,k\}}} \prod_{s=1}^{r} \mathbf{1}_{\{i_{g_{2s-1}}=\ i_{g_{2s}}\neq 0\}}\times$$

$$\times\hat{J}^S\left[\tilde{T}^{k-2r}_{g_1,g_2\ldots,g_{2r-1},g_{2r}}\Phi\right]_{T,t}^{(i_{q_1}\ldots i_{q_{k-2r}})} \quad \text{w. p. 1.} \tag{41}$$

It is interesting to compare the formulas (34) and (41) with similar formulas (12) and (13) that connect the iterated Stratonovich and Itô stochastic integrals.

Further, we will consider the connection of the above results with Theorems 4 and 7. Suppose that the condition (38) is fulfilled and all limiting traces $\tilde{T}^{k-2r}_{g_1,g_2\ldots,g_{2r-1},g_{2r}}\Phi$ exist for all $r=1,2,\ldots,[k/2]$ and for all possible $g_1,g_2,\ldots,g_{2r-1},g_{2r}$.

Using (32) and (38), we get

$$\mathsf{M}\left\{\left(J'\left[T^{k-2r,p}_{g_1,g_2\ldots,g_{2r-1},g_{2r}}\Phi\right]_{T,t}^{(i_{q_1}\ldots i_{q_{k-2r}})} - J'\left[\tilde{T}^{k-2r,p}_{g_1,g_2\ldots,g_{2r-1},g_{2r}}\Phi\right]_{T,t}^{(i_{q_1}\ldots i_{q_{k-2r}})}\right)^2\right\} \le$$

$$\le K_{k-2r}\left\|T^{k-2r,p}_{g_1,g_2\ldots,g_{2r-1},g_{2r}}\Phi - \tilde{T}^{k-2r,p}_{g_1,g_2\ldots,g_{2r-1},g_{2r}}\Phi\right\|^2_{L_2([t,T]^{k-2r})} \ \to\ 0,$$

$$\mathsf{M}\left\{\left(J'\left[\tilde{T}^{k-2r}_{g_1,g_2\ldots,g_{2r-1},g_{2r}}\Phi\right]_{T,t}^{(i_{q_1}\ldots i_{q_{k-2r}})} - J'\left[\tilde{T}^{k-2r,p}_{g_1,g_2\ldots,g_{2r-1},g_{2r}}\Phi\right]_{T,t}^{(i_{q_1}\ldots i_{q_{k-2r}})}\right)^2\right\} \le$$

$$\le K'_{k-2r}\left\|\tilde{T}^{k-2r}_{g_1,g_2\ldots,g_{2r-1},g_{2r}}\Phi - \tilde{T}^{k-2r,p}_{g_1,g_2\ldots,g_{2r-1},g_{2r}}\Phi\right\|^2_{L_2([t,T]^{k-2r})} \ \to\ 0$$

if $p\to\infty$ (for all $r=1,2,\ldots,[k/2]$ and for all possible $g_1,g_2,\ldots,g_{2r-1},g_{2r}$ (see (5))), where K_{k-2r}, K'_{k-2r} are constants. Then w. p. 1

$$\underset{p\to\infty}{\text{l.i.m.}}\ J'\left[T^{k-2r,p}_{g_1,g_2\ldots,g_{2r-1},g_{2r}}\Phi\right]_{T,t}^{(i_{q_1}\ldots i_{q_{k-2r}})} = \underset{p\to\infty}{\text{l.i.m.}}\ J'\left[\tilde{T}^{k-2r,p}_{g_1,g_2\ldots,g_{2r-1},g_{2r}}\Phi\right]_{T,t}^{(i_{q_1}\ldots i_{q_{k-2r}})} =$$

$$= J'\left[\tilde{T}^{k-2r}_{g_1,g_2\ldots,g_{2r-1},g_{2r}}\Phi\right]_{T,t}^{(i_{q_1}\ldots i_{q_{k-2r}})}. \tag{42}$$

Passing to the limit $\underset{p\to\infty}{\text{l.i.m.}}$ in (31) and applying (42), we obtain w. p. 1

$$\hat{J}^S[\Phi]_{T,t}^{(i_1\ldots i_k)} = \underset{p\to\infty}{\text{l.i.m.}}\ \hat{J}_p^S[\Phi]_{T,t}^{(i_1\ldots i_k)} = \underset{p\to\infty}{\text{l.i.m.}}\ J'[\Phi_p]_{T,t}^{(i_1\ldots i_k)}+$$

$$+\sum_{r=1}^{[k/2]}\sum_{\substack{(\{\{g_1,g_2\},\ldots,\{g_{2r-1},g_{2r}\}\},\{q_1,\ldots,q_{k-2r}\})\\ \{g_1,g_2,\ldots,g_{2r-1},g_{2r},q_1,\ldots,q_{k-2r}\}=\{1,2,\ldots,k\}}}\prod_{s=1}^{r}\mathbf{1}_{\{i_{g_{2s-1}}=\ i_{g_{2s}}\neq 0\}}\times$$

$$\times\underset{p\to\infty}{\text{l.i.m.}}\ J'\Biggl[\tilde{T}^{k-2r,p}_{g_1,g_2\ldots,g_{2r-1},g_{2r}}\Phi\Biggr]_{T,t}^{(i_{q_1}\ldots i_{q_{k-2r}})}, \tag{43}$$

where notations are the same as in (31).

Replacing the function $\Phi(t_1,\ldots,t_k)$ in (43) with the Volterra-type kernel (3), where $\psi_1(\tau),\ldots,\psi_k(\tau)\in L_2([t,T])$, and using Theorem 8, we have

$$\underset{p\to\infty}{\text{l.i.m.}}\sum_{j_1,\ldots,j_k=0}^{p}C_{j_k\ldots j_1}\prod_{l=1}^{k}\zeta_{j_l}^{(i_l)} = J[\psi^{(k)}]_{T,t}^{(i_1\ldots i_k)}+$$

$$+\sum_{r=1}^{[k/2]}\sum_{\substack{(\{\{g_1,g_2\},\ldots,\{g_{2r-1},g_{2r}\}\},\{q_1,\ldots,q_{k-2r}\})\\ \{g_1,g_2,\ldots,g_{2r-1},g_{2r},q_1,\ldots,q_{k-2r}\}=\{1,2,\ldots,k\}}}\prod_{s=1}^{r}\mathbf{1}_{\{i_{g_{2s-1}}=\ i_{g_{2s}}\neq 0\}}\times$$

$$\times\underset{p\to\infty}{\text{l.i.m.}}\ J'\Biggl[\tilde{T}^{k-2r,p}_{g_1,g_2\ldots,g_{2r-1},g_{2r}}K\Biggr]_{T,t}^{(i_{q_1}\ldots i_{q_{k-2r}})}, \tag{44}$$

where $J[\psi^{(k)}]_{T,t}^{(i_1\ldots i_k)}$ is the iterated Itô stochastic integral (9).

Consider the equality (see the proof of Theorem 2.49 in [16], Sect. 2.22)

$$\underset{p\to\infty}{\text{l.i.m.}}\sum_{j_{q_1},\ldots,j_{q_{k-2r}}=0}^{p}\frac{1}{2^r}C_{j_k\ldots j_1}\Biggl|_{(j_{g_2}j_{g_1})\curvearrowright(\cdot)\ldots(j_{g_{2r}}j_{g_{2r-1}})\curvearrowright(\cdot),j_{g_1}=\ j_{g_2},\ldots,j_{g_{2r-1}}=\ j_{g_{2r}}}\times$$

$$\times\prod_{s=1}^{r}\mathbf{1}_{\{i_{g_{2s-1}}=\ i_{g_{2s}}\neq 0\}}J'[\phi_{j_{q_1}}\ldots\phi_{j_{q_{k-2r}}}]_{T,t}^{(i_{q_1}\ldots i_{q_{k-2r}})} =$$

$$= \frac{1}{2^r} J[\psi^{(k)}]_{T,t}^{(i_1\ldots i_k)[s_r,\ldots,s_1]} \tag{45}$$

w. p. 1, where $g_2 = g_1 + 1, \ldots, g_{2r} = g_{2r-1} + 1$, $g_{2i-1} \stackrel{\mathsf{def}}{=} s_i$; $i = 1, 2, \ldots, r$; $r = 1, 2, \ldots, [k/2]$, $(s_r, \ldots, s_1) \in \mathrm{A}_{k,r}$, $J[\psi^{(k)}]_{T,t}^{(i_1\ldots i_k)[s_r,\ldots,s_1]}$ is defined by (7) and $\mathrm{A}_{k,r}$ is defined by (8); another notations in (45) are the same as in Sect. 2.1.

Also, consider the equality (see the proof in [16], Sect. 2.27.4 and 2.30)

$$\lim_{p\to\infty} \sum_{j_{g_1}, j_{g_3}, \ldots, j_{g_{2r-1}}=0}^{p} C_{j_k\ldots j_1}\Bigl|_{j_{g_1}=j_{g_2},\ldots,j_{g_{2r-1}}=j_{g_{2r}}} =$$

$$= \frac{1}{2^r} \prod_{l=1}^{r} \mathbf{1}_{\{g_{2l}=g_{2l-1}+1\}} C_{j_k\ldots j_1}\Bigl|_{(j_{g_2} j_{g_1})\curvearrowright(\cdot)\ldots(j_{g_{2r}} j_{g_{2r-1}})\curvearrowright(\cdot), j_{g_1}=\, j_{g_2},\ldots,j_{g_{2r-1}}=\, j_{g_{2r}}} \tag{46}$$

for all possible $g_1, g_2, \ldots, g_{2r-1}, g_{2r}$ (see (5)), where $k \geq 2r$, $r = 1, 2, \ldots, [k/2]$, $C_{j_k\ldots j_1}$ is defined by (2); another notations are the same as in Sect. 2.1.

Note that

$$\sum_{\substack{(\{\{g_1,g_2\},\ldots,\{g_{2r-1},g_{2r}\}\},\{q_1,\ldots,q_{k-2r}\}) \\ \{g_1,g_2,\ldots,g_{2r-1},g_{2r},q_1,\ldots,q_{k-2r}\}=\{1,2,\ldots,k\}}} \Bigl|_{g_2=g_1+1, g_3=g_2+1,\ldots,g_{2r}=g_{2r-1}+1} A_{g_1,g_3,\ldots,g_{2r-1}} =$$

$$= \sum_{(s_r,\ldots,s_1)\in \mathrm{A}_{k,r}} A_{s_1,s_2,\ldots,s_r}, \tag{47}$$

where $A_{g_1,g_3,\ldots,g_{2r-1}}$, $A_{s_1,s_2,\ldots,s_r}$ are scalar values, $g_{2i-1} = s_i$; $i = 1, 2, \ldots, r$; $r = 1, 2, \ldots, [k/2]$, $\mathrm{A}_{k,r}$ is defined by (8).

Combining (35), (36), (45), (46) and (47), we obtain w. p. 1

$$\prod_{s=1}^{r} \mathbf{1}_{\{i_{g_{2s-1}}=\, i_{g_{2s}}\neq 0\}} \underset{p\to\infty}{\text{l.i.m.}}\ J'\Bigl[\tilde{T}^{k-2r,p}_{g_1,g_2\ldots,g_{2r-1},g_{2r}} K\Bigr]_{T,t}^{(i_{q_1}\ldots i_{q_{k-2r}})} =$$

$$= \prod_{l=1}^{r} \mathbf{1}_{\{g_{2l}=g_{2l-1}+1\}} \frac{1}{2^r} J[\psi^{(k)}]_{T,t}^{(i_1\ldots i_k)[s_r,\ldots,s_1]},$$

and

$$J[\psi^{(k)}]_{T,t}^{(i_1\ldots i_k)}+$$

$$+\sum_{r=1}^{[k/2]}\sum_{\substack{(\{\{g_1,g_2\},\ldots,\{g_{2r-1},g_{2r}\}\},\{q_1,\ldots,q_{k-2r}\})\\ \{g_1,g_2,\ldots,g_{2r-1},g_{2r},q_1,\ldots,q_{k-2r}\}=\{1,2,\ldots,k\}}}\prod_{s=1}^{r}\mathbf{1}_{\{i_{g_{2s-1}}=\ i_{g_{2s}}\neq 0\}}\times$$

$$\times\underset{p\to\infty}{\text{l.i.m.}}\ J'\left[\tilde{T}^{k-2r,p}_{g_1,g_2\ldots,g_{2r-1},g_{2r}}K\right]_{T,t}^{(i_{q_1}\ldots i_{q_{k-2r}})}=$$

$$=J[\psi^{(k)}]_{T,t}^{(i_1\ldots i_k)}+\sum_{r=1}^{[k/2]}\frac{1}{2^r}\sum_{(s_r,\ldots,s_1)\in\mathrm{A}_{k,r}}J[\psi^{(k)}]_{T,t}^{(i_1\ldots i_k)[s_r,\ldots,s_1]}. \tag{48}$$

Suppose that $\psi_1(\tau),\ldots,\psi_k(\tau)$ are continuous functions on $[t,T]$. Further, applying Theorem 5 to the right-hand side of (48) and combining (44) and (48), we get the following expansion:

$$J^*[\psi^{(k)}]_{T,t}^{(i_1\ldots i_k)}=\underset{p\to\infty}{\text{l.i.m.}}\sum_{j_1,\ldots,j_k=0}^{p}C_{j_k\ldots j_1}\prod_{l=1}^{k}\zeta_{j_l}^{(i_l)},$$

where $J^*[\psi^{(k)}]_{T,t}^{(i_1\ldots i_k)}$ is the iterated Stratonovich stochastic integral (1).

Note that in our case (the case of the Volterra-type kernel (3)), limiting traces $\tilde{T}^{k-2r}_{g_1,g_2\ldots,g_{2r-1},g_{2r}}K$ exist for all $r=1,2,\ldots,[k/2]$ and for all possible $g_1,g_2,\ldots,g_{2r-1},g_{2r}$, and are determined by the equality (23) (also see (19)). Moreover, the condition (38) in this case has the form (11) (or (24)).

Further, we will consider another approach to deriving the Hu–Meyer formulas based on the so-called Hilbert space valued traces [2], [3], [5]. At that we will still consider the case of a multidimensional Wiener process.

Consider the following definition of multiple Stratonovich stochastic integral (see, for example, [3]):

$$\underset{N\to\infty}{\text{l.i.m.}}\sum_{j_1,\ldots,j_k=0}^{N-1}\Phi\left(\tau_{j_1},\ldots,\tau_{j_k}\right)\prod_{l=1}^{k}\Delta\mathbf{w}_{\tau_{j_l}}^{(i_l)}\stackrel{\text{def}}{=}J[\Phi]_{T,t}^{(i_1\ldots i_k)}, \tag{49}$$

where we suppose that $\Phi(t_1,\ldots,t_k):[t,T]^k\to\mathbf{R}$ is a continuous nonrandom function, $i_1,\ldots,i_k=0,1,\ldots,m$, $\Delta\mathbf{w}_{\tau_j}^{(i)}=\mathbf{w}_{\tau_{j+1}}^{(i)}-\mathbf{w}_{\tau_j}^{(i)}$ $(i=1,\ldots,m)$, $\Delta\mathbf{w}_{\tau_j}^{(0)}=\tau_{j+1}-\tau_j$, $\{\tau_j\}_{j=0}^{N}$ is a partition of $[t,T]$ which satisfies the following condition:

$$t=\tau_0<\tau_1<\ldots<\tau_N=T,\quad \max_{0\le j\le N-1}|\tau_{j+1}-\tau_j|\to 0\ \text{ if }\ N\to\infty. \tag{50}$$

Note that the function $\Phi(t_1,\ldots,t_k)$ in (49) can be discontinuous. For example, it can have the form (2.379) (see [16], Sect. 2.4.1).

Further, let us write the formula (2.429) (see [16], Sect. 2.4.1) for the function $\Phi(t_1,\ldots,t_4)$ and use the formula (1.329) (see [16], Sect. 1.11) for the multiple Wiener stochastic integral $J'[\Phi]_{T,t}^{(i_1\ldots i_4)}$ defined by (1.328) (see [16], Sect. 1.11)

$$J[\Phi]_{T,t}^{(i_1\ldots i_4)}=J'[\Phi]_{T,t}^{(i_1\ldots i_4)}+$$

$$+\mathbf{1}_{\{i_1=i_2\ne 0\}}J'\left[\Phi(t_1,\ldots,t_4)\Big|_{t_1=t_2}\right]_{T,t}^{(0i_3i_4)}+\mathbf{1}_{\{i_1=i_3\ne 0\}}J'\left[\Phi(t_1,\ldots,t_4)\Big|_{t_1=t_3}\right]_{T,t}^{(0i_2i_4)}+$$

$$+\mathbf{1}_{\{i_1=i_4\ne 0\}}J'\left[\Phi(t_1,\ldots,t_4)\Big|_{t_1=t_4}\right]_{T,t}^{(0i_2i_3)}+\mathbf{1}_{\{i_2=i_3\ne 0\}}J'\left[\Phi(t_1,\ldots,t_4)\Big|_{t_2=t_3}\right]_{T,t}^{(i_10i_4)}+$$

$$+\mathbf{1}_{\{i_2=i_4\ne 0\}}J'\left[\Phi(t_1,\ldots,t_4)\Big|_{t_2=t_4}\right]_{T,t}^{(i_10i_3)}+\mathbf{1}_{\{i_3=i_4\ne 0\}}J'\left[\Phi(t_1,\ldots,t_4)\Big|_{t_3=t_4}\right]_{T,t}^{(i_1i_20)}+$$

$$+\mathbf{1}_{\{i_1=i_2\ne 0\}}\mathbf{1}_{\{i_3=i_4\ne 0\}}J'\left[\Phi(t_1,\ldots,t_4)\Big|_{t_1=t_2,t_3=t_4}\right]_{T,t}^{(00)}+$$

$$+\mathbf{1}_{\{i_1=i_3\ne 0\}}\mathbf{1}_{\{i_2=i_4\ne 0\}}J'\left[\Phi(t_1,\ldots,t_4)\Big|_{t_1=t_3,t_2=t_4}\right]_{T,t}^{(00)}+$$

$$+\mathbf{1}_{\{i_1=i_4\ne 0\}}\mathbf{1}_{\{i_2=i_3\ne 0\}}J'\left[\Phi(t_1,\ldots,t_4)\Big|_{t_1=t_4,t_2=t_3}\right]_{T,t}^{(00)} \tag{51}$$

w. p. 1. Applying a Fubini-type theorem for stochastic and Lebesgue intgrals (as in [7]) to the right-hand side of (51), we obtain w. p. 1

$$J[\Phi]_{T,t}^{(i_1\ldots i_4)}=J'[\Phi]_{T,t}^{(i_1\ldots i_4)}+\mathbf{1}_{\{i_1=i_2\ne 0\}}J'\left[\int\limits_{[t,T]}\Phi(t_1,\ldots,t_4)\Big|_{t_1=t_2}dt_1\right]_{T,t}^{(i_3i_4)}+$$

$$+\mathbf{1}_{\{i_1=i_3\neq 0\}}J'\left[\int\limits_{[t,T]}\Phi(t_1,\ldots,t_4)\Big|_{t_1=t_3}dt_1\right]_{T,t}^{(i_2i_4)}+$$

$$+\mathbf{1}_{\{i_1=i_4\neq 0\}}J'\left[\int\limits_{[t,T]}\Phi(t_1,\ldots,t_4)\Big|_{t_1=t_4}dt_1\right]_{T,t}^{(i_2i_3)}+$$

$$+\mathbf{1}_{\{i_2=i_3\neq 0\}}J'\left[\int\limits_{[t,T]}\Phi(t_1,\ldots,t_4)\Big|_{t_2=t_3}dt_2\right]_{T,t}^{(i_1i_4)}+$$

$$+\mathbf{1}_{\{i_2=i_4\neq 0\}}J'\left[\int\limits_{[t,T]}\Phi(t_1,\ldots,t_4)\Big|_{t_2=t_4}dt_2\right]_{T,t}^{(i_1i_3)}+$$

$$+\mathbf{1}_{\{i_3=i_4\neq 0\}}J'\left[\int\limits_{[t,T]}\Phi(t_1,\ldots,t_4)\Big|_{t_3=t_4}dt_3\right]_{T,t}^{(i_1i_2)}+$$

$$+\mathbf{1}_{\{i_1=i_2\neq 0\}}\mathbf{1}_{\{i_3=i_4\neq 0\}}\int\limits_{[t,T]^2}\Phi(t_1,\ldots,t_4)\Big|_{t_1=t_2,t_3=t_4}dt_1dt_3+$$

$$+\mathbf{1}_{\{i_1=i_3\neq 0\}}\mathbf{1}_{\{i_2=i_4\neq 0\}}\int\limits_{[t,T]^2}\Phi(t_1,\ldots,t_4)\Big|_{t_1=t_3,t_2=t_4}dt_1dt_2+$$

$$+\mathbf{1}_{\{i_1=i_4\neq 0\}}\mathbf{1}_{\{i_2=i_3\neq 0\}}\int\limits_{[t,T]^2}\Phi(t_1,\ldots,t_4)\Big|_{t_1=t_4,t_2=t_3}dt_1dt_2. \qquad (52)$$

Let us generalize the formula (52) to the case $k\in\mathbf{N}$. As a result, we obtain the following Hu–Meyer formula for the case of a multidimensional Wiener process:

$$J[\Phi]_{T,t}^{(i_1\ldots i_k)}=J'[\Phi]_{T,t}^{(i_1\ldots i_k)}+$$

$$+\sum_{r=1}^{[k/2]}\sum_{\substack{(\{\{g_1,g_2\},\ldots,\{g_{2r-1},g_{2r}\}\},\{q_1,\ldots,q_{k-2r}\})\\ \{g_1,g_2,\ldots,g_{2r-1},g_{2r},q_1,\ldots,q_{k-2r}\}=\{1,2,\ldots,k\}}}\prod_{s=1}^{r}\mathbf{1}_{\{i_{g_{2s-1}}=\ i_{g_{2s}}\neq 0\}}\times$$

$$\times J'\left[\breve{T}^{k-2r}_{g_1,g_2\ldots,g_{2r-1},g_{2r}}\Phi\right]_{T,t}^{(i_{q_1}\ldots i_{q_{k-2r}})}\quad \text{w. p. 1}, \qquad (53)$$

where the so-called Hilbert space valued trace $\breve{T}^{k-2r}_{g_1,g_2\ldots,g_{2r-1},g_{2r}}\Phi \in L_2([t,T]^{k-2r})$ has the form (we assume that this trace exists)

$$\breve{T}^{k-2r}_{g_1,g_2\ldots,g_{2r-1},g_{2r}}\Phi(t_{q_1},\ldots,t_{q_{k-2r}}) \stackrel{\sf def}{=}$$

$$\stackrel{\sf def}{=} \int\limits_{[t,T]^r} \Phi(t_1,\ldots,t_k)\Biggl|_{t_{g_1}=t_{g_2},\ldots,t_{g_{2r-1}}=t_{g_{2r}}} dt_{g_1}\ldots dt_{g_{2r-1}},$$

where $J'\Biggl[\breve{T}^{k-2r}_{g_1,g_2\ldots,g_{2r-1},g_{2r}}\Phi\Biggr]^{(i_{q_1}\ldots i_{q_{k-2r}})}_{T,t} \stackrel{\sf def}{=} \breve{T}^{k-2r}_{g_1,g_2\ldots,g_{2r-1},g_{2r}}\Phi$ for $k=2r$; another notations are the same as in (34). Recall that in [3], the following particular case $\breve{T}^{k-2r}_{1,2\ldots,2r-1,2r}\Phi(t_{2r+1},\ldots,t_k)$ of $\breve{T}^{k-2r}_{g_1,g_2\ldots,g_{2r-1},g_{2r}}\Phi(t_{q_1},\ldots,t_{q_{k-2r}})$ was considered.

The inverse version of the Hu–Meyer formula (53) has the form

$$J'[\Phi]^{(i_1\ldots i_k)}_{T,t} = J[\Phi]^{(i_1\ldots i_k)}_{T,t}+$$

$$+\sum_{r=1}^{[k/2]}(-1)^r \sum_{\substack{(\{\{g_1,g_2\},\ldots,\{g_{2r-1},g_{2r}\}\},\{q_1,\ldots,q_{k-2r}\})\\ \{g_1,g_2,\ldots,g_{2r-1},g_{2r},q_1,\ldots,q_{k-2r}\}=\{1,2,\ldots,k\}}} \prod_{s=1}^{r}\mathbf{1}_{\{i_{g_{2s-1}}=\ i_{g_{2s}}\ne 0\}}\times$$

$$\times J\Biggl[\breve{T}^{k-2r}_{g_1,g_2\ldots,g_{2r-1},g_{2r}}\Phi\Biggr]^{(i_{q_1}\ldots i_{q_{k-2r}})}_{T,t} \quad \text{w. p. 1.}$$

Note that there are more general definitions of the multiple Stratonovich stochastic integral (see, for example, (1.5.9) and Sect. 2.1 in [3]) that are consistent with definition (49) on the class of continuous functions.

Further, consider one of these generalizations. But first, consider the following Riemann–Stieltjes integral:

$$\int\limits_{[t,T]^k} \Phi(t_1,\ldots,t_k)d\mathbf{w}^{(i_1)p}_{t_1}\ldots d\mathbf{w}^{(i_k)p}_{t_k}, \tag{54}$$

where $p\in\mathbf{N}$, $i_1,\ldots,i_k=0,1,\ldots,m$, $\Phi(t_1,\ldots,t_k):\ [t,T]^k\to\mathbf{R}$ is a nonrandom function, $\mathbf{w}_\tau$ is a random vector with an $m+1$ components ($\mathbf{w}^{(i)}_\tau$ $(i=1,\ldots,m)$

are independent standard Wiener processes and $\mathbf{w}_{\tau}^{(0)} = \tau$), $\mathbf{w}_{\tau}^{(i)p}$ is the following mean-square approximation of $\mathbf{w}_{\tau}^{(i)}$ [42]:

$$\mathbf{w}_{\tau}^{(i)p} = \mathbf{w}_{t}^{(i)p} + \sum_{j=0}^{p} \int\limits_{t}^{\tau} \phi_j(s)ds\ \zeta_j^{(i)}, \tag{55}$$

where

$$\zeta_j^{(i)} = \int\limits_{t}^{T} \phi_j(s)d\mathbf{w}_s^{(i)},$$

$\tau \in [t,T]$, $t \geq 0$, $\{\phi_j(x)\}_{j=0}^{\infty}$ is an arbitrary CONS in $L_2([t,T])$, and $\zeta_j^{(i)}$ are independent standard Gaussian random variables for various i or j (in the case $i \neq 0$).

From (55) we have

$$d\mathbf{w}_{\tau}^{(i)p} = \sum_{j=0}^{p} \phi_j(\tau)\zeta_j^{(i)}d\tau. \tag{56}$$

Let us substitute (56) into (54)

$$\int\limits_{[t,T]^k} \Phi(t_1,\ldots,t_k)d\mathbf{w}_{t_1}^{(i_1)p}\ldots d\mathbf{w}_{t_k}^{(i_k)p} = \sum_{j_1,\ldots,j_k=0}^{p} C_{j_k\ldots j_1}\prod_{l=1}^{k}\zeta_{j_l}^{(i_l)}, \tag{57}$$

where

$$C_{j_k\ldots j_1} = \int\limits_{[t,T]^k} \Phi(t_1,\ldots,t_k)\prod_{l=1}^{k}\phi_{j_l}(t_l)dt_1\ldots dt_k$$

is the Fourier coefficient.

The formula (57) explains why $\hat{J}^S[\Phi]_{T,t}^{(i_1\ldots i_k)}$ (see (29), (30)) was called the multiple Stratonovich stochastic integral.

Now consider another approximation of the Wiener process, namely the so-called polygonal approximation (see [2], Example 5.17)

$$\mathbf{w}_{\tau}^{(i)N} = \sum_{j=0}^{N-1}\left(\mathbf{w}_{\tau_j}^{(i)} + \frac{1}{\Delta_j}\Delta\mathbf{w}_{\tau_j}^{(i)}(\tau-\tau_j)\right)\mathbf{1}_{T_j}(\tau), \tag{58}$$

where $N \in \mathbf{N}$, $i = 0, 1, \ldots, m$, $\tau \in [t, T]$, $t \geq 0$, $\mathbf{w}_\tau$ is a random vector with an $m+1$ components ($\mathbf{w}_\tau^{(i)}$ $(i = 1, \ldots, m)$ are independent standard Wiener processes and $\mathbf{w}_\tau^{(0)} = \tau$), $\Delta\mathbf{w}_{\tau_j}^{(i)} = \mathbf{w}_{\tau_{j+1}}^{(i)} - \mathbf{w}_{\tau_j}^{(i)}$ $(i = 1, \ldots, m)$, $\Delta\mathbf{w}_{\tau_j}^{(0)} = \Delta_j$, $\Delta_j = \tau_{j+1} - \tau_j$, $T_j = [\tau_j, \tau_{j+1})$, $\mathbf{1}_A$ is the indicator of the set A, $\{\tau_j\}_{j=0}^N$ is a partition of the interval $[t, T]$ which satisfies the condition (50).

From (58) we have

$$d\mathbf{w}_\tau^{(i)N} = \sum_{j=0}^{N-1} \frac{1}{\Delta_j} \Delta\mathbf{w}_{\tau_j}^{(i)} d\tau \mathbf{1}_{T_j}(\tau). \tag{59}$$

Using (59), we obtain

$$\int\limits_{[t,T]^k} \Phi(t_1, \ldots, t_k) d\mathbf{w}_{t_1}^{(i_1)N} \ldots d\mathbf{w}_{t_k}^{(i_k)N} =$$

$$= \sum_{j_1,\ldots,j_k=0}^{N-1} \frac{1}{\Delta_{j_1} \ldots \Delta_{j_k}} \times$$

$$\times \int\limits_{[t,T]^k} \Phi(t_1, \ldots, t_k) \mathbf{1}_{T_{j_1}}(t_1) \ldots \mathbf{1}_{T_{j_k}}(t_k) dt_1 \ldots dt_k \Delta\mathbf{w}_{\tau_{j_1}}^{(i_1)} \ldots \Delta\mathbf{w}_{\tau_{j_k}}^{(i_k)} =$$

$$= \sum_{j_1,\ldots,j_k=0}^{N-1} \frac{1}{\Delta_{j_1} \ldots \Delta_{j_k}} \int\limits_{T_{j_1} \times \ldots \times T_{j_k}} \Phi(t_1, \ldots, t_k) dt_1 \ldots dt_k \Delta\mathbf{w}_{\tau_{j_1}}^{(i_1)} \ldots \Delta\mathbf{w}_{\tau_{j_k}}^{(i_k)}.$$

We define the multiple Stratonovich stochastic integral (see [2], [43]) for $\Phi(t_1, \ldots, t_k) \in L_2([t, T]^k)$ (but this function satisfies an additional condition (see below)) as the following mean-square limit (the case of a multidimensional Wiener process):

$$\bar{J}^S[\Phi]_{T,t}^{(i_1 \ldots i_k)} =$$

$$= \underset{N \to \infty}{\text{l.i.m.}} \sum_{j_1,\ldots,j_k=0}^{N-1} \frac{1}{\Delta_{j_1} \ldots \Delta_{j_k}} \int\limits_{T_{j_1} \times \ldots \times T_{j_k}} \Phi(t_1, \ldots, t_k) dt_1 \ldots dt_k \Delta\mathbf{w}_{\tau_{j_1}}^{(i_1)} \ldots \Delta\mathbf{w}_{\tau_{j_k}}^{(i_k)}, \tag{60}$$

where $i_1,\ldots,i_k=0,1,\ldots,m$ and $\{\tau_j\}_{j=0}^{N}$ as in (50).

We define the rth trace of the function $\Phi(t_1,\ldots,t_k)\in L_2([t,T]^k)$ by the following expression [2], [43]:

$$\bar{T}^{k-2r}_{g_1,g_2\ldots,g_{2r-1},g_{2r}}\Phi(t_{q_1},\ldots,t_{q_{k-2r}})\stackrel{\sf def}{=}$$

$$\stackrel{\sf def}{=}\lim_{N\to\infty}\bar{T}^{k-2r,N}_{g_1,g_2\ldots,g_{2r-1},g_{2r}}\Phi(t_{q_1},\ldots,t_{q_{k-2r}}) \tag{61}$$

in $L_2([t,T]^{k-2r})$, where

$$\bar{T}^{k-2r,N}_{g_1,g_2\ldots,g_{2r-1},g_{2r}}\Phi(t_{q_1},\ldots,t_{q_{k-2r}})=\sum_{j_{g_1},j_{g_3},\ldots,j_{g_{2r-1}}=0}^{N-1}\frac{1}{\Delta_{j_{g_1}}\Delta_{j_{g_3}},\ldots\Delta_{j_{g_{2r-1}}}}\times$$

$$\times\int\limits_{T^2_{j_{g_1}}\times T^2_{j_{g_3}}\times\ldots\times T^2_{j_{g_{2r-1}}}}\Phi(t_1,\ldots,t_k)dt_{g_1}dt_{g_2}\ldots dt_{g_{2r-1}}dt_{g_{2r}},$$

$\{g_1,g_2,\ldots,g_{2r-1},g_{2r},q_1,\ldots,q_{k-2r}\}=\{1,2,\ldots,k\}$ (see (5)), $r=1,2,\ldots,[k/2]$.

Assume that all traces $\bar{T}^{k-2r}_{g_1,g_2\ldots,g_{2r-1},g_{2r}}\Phi(t_{q_1},\ldots,t_{q_{k-2r}})$ (for all $r=1,2,\ldots,[k/2]$ and for all possible $g_1,g_2,\ldots,g_{2r-1},g_{2r}$ (see (5))) exist.

Replacing $\Phi\left(\tau_{j_1},\ldots,\tau_{j_k}\right)$ with

$$\frac{1}{\Delta_{j_1}\ldots\Delta_{j_k}}\int\limits_{T_{j_1}\times\ldots\times T_{j_k}}\Phi(t_1,\ldots,t_k)dt_1\ldots dt_k$$

in deriving the formula (53), we get the following Hu–Meyer formula for the multiple Stratonovich stochastic integral (60):

$$\bar{J}^S[\Phi]_{T,t}^{(i_1\ldots i_k)}=J'[\Phi]_{T,t}^{(i_1\ldots i_k)}+$$

$$+\sum_{r=1}^{[k/2]}\sum_{\substack{(\{\{g_1,g_2\},\ldots,\{g_{2r-1},g_{2r}\}\},\{q_1,\ldots,q_{k-2r}\})\\ \{g_1,g_2,\ldots,g_{2r-1},g_{2r},q_1,\ldots,q_{k-2r}\}=\{1,2,\ldots,k\}}}\prod_{s=1}^{r}\mathbf{1}_{\{i_{g_{2s-1}}=\ i_{g_{2s}}\neq 0\}}\times$$

$$\times J'\left[\bar{T}^{k-2r}_{g_1,g_2\ldots,g_{2r-1},g_{2r}}\Phi\right]_{T,t}^{(i_{q_1}\ldots i_{q_{k-2r}})}\quad\text{w. p. 1,} \tag{62}$$

where $J'\bigl[\bar{T}^{k-2r}_{g_1,g_2\ldots,g_{2r-1},g_{2r}}\Phi\bigr]^{(i_{q_1}\ldots i_{q_{k-2r}})}_{T,t} \stackrel{\sf def}{=} \bar{T}^{k-2r}_{g_1,g_2\ldots,g_{2r-1},g_{2r}}\Phi$ for $k=2r$, the existence of the right-hand side of (62) ensures the existence of the multiple Stratonovich stochastic integral $\bar{J}^S[\Phi]^{(i_1\ldots i_k)}_{T,t}$; another notations are the same as in (34).

Assume that all limiting traces $T^{k-2r}_{g_1,\ldots,g_{2r}}\Phi(t_{q_1},\ldots,t_{q_{k-2r}})$ (see (26)) exist for all $r=1,2,\ldots,[k/2]$ and for all possible $g_1,g_2,\ldots,g_{2r-1},g_{2r}$ (see (5)).

Moreover, suppose that

$$\bar{T}^{k-2r}_{g_1,g_2\ldots,g_{2r-1},g_{2r}}\Phi(t_{q_1},\ldots,t_{q_{k-2r}}) =$$

$$= T^{k-2r}_{g_1,g_2\ldots,g_{2r-1},g_{2r}}\Phi(t_{q_1},\ldots,t_{q_{k-2r}})$$

almost everywhere on $[t,T]^{k-2r}$ (with respect to Lebesgue's measure) for all $r=1,2,\ldots,[k/2]$ and for all possible $g_1,g_2,\ldots,g_{2r-1},g_{2r}$ (see (5)), where $\bar{T}^{k-2r}_{g_1,g_2\ldots,g_{2r-1},g_{2r}}\Phi(t_{q_1},\ldots,t_{q_{k-2r}})$ is defined by (61).

Then, using (34), (62), and (32), we obtain

$$\mathsf{M}\left\{\left(\bar{J}^S[\Phi]^{(i_1\ldots i_k)}_{T,t}-\hat{J}^S[\Phi]^{(i_1\ldots i_k)}_{T,t}\right)^2\right\}\le$$

$$\le \bar{C}_k\sum_{r=1}^{[k/2]}\sum_{\substack{(\{\{g_1,g_2\},\ldots,\{g_{2r-1},g_{2r}\}\},\{q_1,\ldots,q_{k-2r}\})\\ \{g_1,g_2,\ldots,g_{2r-1},g_{2r},q_1,\ldots,q_{k-2r}\}=\{1,2,\ldots,k\}}}\prod_{s=1}^{r}\mathbf{1}_{\{i_{g_{2s-1}}=\ i_{g_{2s}}\ne 0\}}\times$$

$$\times\left\|\bar{T}^{k-2r}_{g_1,g_2\ldots,g_{2r-1},g_{2r}}\Phi-T^{k-2r}_{g_1,g_2\ldots,g_{2r-1},g_{2r}}\Phi\right\|^2_{L_2([t,T]^{k-2r})}=0,$$

where $\bar{C}_k$ is a constant.

Then (see (29), (30))

$$\bar{J}^S[\Phi]^{(i_1\ldots i_k)}_{T,t}=\hat{J}^S[\Phi]^{(i_1\ldots i_k)}_{T,t}=\underset{p\to\infty}{\text{l.i.m.}}\sum_{j_1,\ldots,j_k=0}^{p}C_{j_k\ldots j_1}\prod_{l=1}^{k}\zeta_{j_l}^{(i_l)}$$

w. p. 1, where $C_{j_k\ldots j_1}$ is the Fourier coefficient defined by (27).

3.3 Expansion of Iterated Stratonovich Stochastic Integrals of Multiplicity 7

In [16] (Sect. 2.31), a method for verifying the condition (11) was proposed. In particular, in [16] (Sect. 2.32–2.34), Theorems 2 and 3 were proved by this method. In Sect. 3.3 and 3.4, we will apply the given method to check the condition (11) for $k = 7$ and 8, $\psi_1(\tau), \ldots, \psi_8(\tau) \equiv 1$, and two CONS in $L_2([t,T])$ (the trigonometric Fourier basis and the basis of Legendre polynomials).

Next, we introduce new notations and formulate the condition (63) (see below), from which the condition (11) will follow.

Let $g_1, g_2, \ldots, g_{2r-1}, g_{2r}$ as in (5) and $k > 2r$, $r \geq 1$. Consider $d_1, e_1, \ldots, d_f, e_f, f \in \mathbf{N}$ such that

$$1 \leq d_1 - e_1 + 1 < \ldots < d_1 - 1 < d_1 < \ldots < d_f - e_f + 1 < \ldots < d_f - 1 < d_f \leq k,$$

$$\{g_1, g_2, \ldots, g_{2r-1}, g_{2r}\} =$$
$$= \{d_1 - e_1 + 1, \ldots, d_1 - 1, d_1\} \cup \ldots \cup \{d_f - e_f + 1, \ldots, d_f - 1, d_f\},$$

$$e_1 + e_2 + \ldots + e_f = 2r, \quad \{1, \ldots, k\} \backslash \{g_1, g_2, \ldots, g_{2r-1}, g_{2r}\} = \{q_1, \ldots, q_{k-2r}\}.$$

Consider the following condition:

$$\Biggl| \sum_{j_{g_1}, j_{g_3}, \ldots, j_{g_{2r-1}}=0}^{p} \Bigl(C_{j_{d_f} \ldots j_{d_f - e_f + 1}}^{\psi_{d_f} \ldots \psi_{d_f - e_f + 1}} (t_{d_f+1}, t_{d_f - e_f}) \ldots$$

$$\ldots C_{j_{d_1} \ldots j_{d_1 - e_1 + 1}}^{\psi_{d_1} \ldots \psi_{d_1 - e_1 + 1}} (t_{d_1+1}, t_{d_1 - e_1}) \Bigr) \Biggr|_{j_{g_1} = j_{g_2}, \ldots, j_{g_{2r-1}} = j_{g_{2r}}} \Biggr| \leq K, \tag{63}$$

where constant K does not depend on p and $t_{d_1+1}, t_{d_1 - e_1}, \ldots, t_{d_f+1}, t_{d_f - e_f}$ (here $d_1 - e_1 \geq 1$ and $d_f + 1 \leq k$), $t_{k+1} \stackrel{\mathsf{def}}{=} T$, $t_0 \stackrel{\mathsf{def}}{=} t$,

$$C_{j_m \ldots j_l}^{\psi_m \ldots \psi_l}(s, \tau) = \int\limits_{\tau}^{s} \psi_m(t_m) \phi_{j_m}(t_m) \ldots \int\limits_{\tau}^{t_{l+1}} \psi_l(t_l) \phi_{j_l}(t_l) dt_l \ldots dt_m,$$

where $t \le \tau < s \le T$, $m \ge l$ $(m, l \in \mathbf{N})$; another notations as in Sect. 2.1.

In [16] (Sect. 2.31) it is shown that the condition (63) implies the condition (11). Moreover, the condition (63) can be weakened. More precisely, the condition (63) can be replaced by the following condition:

$$\left(\sum_{j_{g_1}, j_{g_3}, \ldots, j_{g_{2r-1}}=0}^{p} \left(C_{j_{d_f} \ldots j_{d_f-e_f+1}}^{\psi_{d_f} \ldots \psi_{d_f-e_f+1}}(t_{d_f+1}, t_{d_f-e_f}) \ldots\right.\right.$$

$$\left.\left. \ldots C_{j_{d_1} \ldots j_{d_1-e_1+1}}^{\psi_{d_1} \ldots \psi_{d_1-e_1+1}}(t_{d_1+1}, t_{d_1-e_1})\right)\Bigg|_{j_{g_1}=j_{g_2}, \ldots, j_{g_{2r-1}}=j_{g_{2r}}}\right)^2 \le F(t_{q_1}, \ldots, t_{q_{k-2r}}) \quad (64)$$

almost everywhere on $X = \{(t_{q_1}, \ldots, t_{q_{k-2r}}) : t \le t_{q_1} \le \ldots \le t_{q_{k-2r}} \le T\}$ with respect to Lebesgue's measure, where the function $F(t_{q_1}, \ldots, t_{q_{k-2r}})$ is such that $\psi_{q_1}^2(t_{q_1}) \ldots \psi_{q_{k-2r}}^2(t_{q_{k-2r}}) F(t_{q_1}, \ldots, t_{q_{k-2r}}) \in L_1([t, T]^{k-2r})$ (see the proof of Theorem 2.60 in [16] and the remark at the end of Sect. 2.31 in [16]).

In this section, we will prove the following theorem.

Theorem 9 [16] (Sect. 2.36). *Suppose that $\{\phi_j(x)\}_{j=0}^{\infty}$ is a CONS of Legendre polynomials or trigonometric Fourier basis in $L_2([t, T])$. Then, for the iterated Stratonovich stochastic integral of seventh multiplicity*

$$J^*[\psi^{(7)}]_{T,t}^{(i_1 \ldots i_7)} = \int\limits_t^{*T} \ldots \int\limits_t^{*t_2} d\mathbf{w}_{t_1}^{(i_1)} \ldots d\mathbf{w}_{t_7}^{(i_7)}$$

the following expansion:

$$J^*[\psi^{(7)}]_{T,t}^{(i_1 \ldots i_7)} = \underset{p \to \infty}{\text{l.i.m.}} \sum_{j_1, \ldots, j_7=0}^{p} C_{j_7 \ldots j_1} \zeta_{j_1}^{(i_1)} \ldots \zeta_{j_7}^{(i_7)}$$

that converges in the mean-square sense is valid, where $i_1, \ldots, i_7 = 0, 1, \ldots, m$,

$$C_{j_7 \ldots j_1} = \int\limits_t^T \phi_{j_7}(t_7) \ldots \int\limits_t^{t_2} \phi_{j_1}(t_1) dt_1 \ldots dt_7,$$

and

$$\zeta_j^{(i)} = \int\limits_t^T \phi_j(\tau) d\mathbf{w}_\tau^{(i)}$$

are independent standard Gaussian random variables for various i or j (in the case $i \neq 0$), $\mathbf{w}_\tau^{(0)} = \tau$.

Proof. In this prove, $k = 7$, $r = 1, 2, 3$, and $\psi_1(\tau), \ldots, \psi_7(\tau) \equiv 1$. For the trigonometric case, we will prove (63). For the polynomial case, we will prove (63) for some cases and prove (64) for remaining cases.

Let throughout this proof

$$C_{j_k \ldots j_1}(s, \tau) = \int\limits_{\tau}^{s} \phi_{j_k}(t_k) \ldots \int\limits_{\tau}^{t_2} \phi_{j_1}(t_1) dt_1 \ldots dt_k,$$

where $k \in \mathbf{N}$ and $t \leq \tau < s \leq T$.

Obviously, that the conditions

$$\left| \sum_{j_1=0}^{p} C_{j_1 j_1}(s, \tau) \right| \leq K, \tag{65}$$

$$\left| \sum_{j_1=0}^{p} C_{j_1}(s, \tau) C_{j_1}(\theta, u) \right| \leq K, \tag{66}$$

$$\left| \sum_{j_1, j_2=0}^{p} C_{j_2 j_2 j_1 j_1}(s, \tau) \right| \leq K, \tag{67}$$

$$\left| \sum_{j_1, j_2=0}^{p} C_{j_2 j_1 j_2 j_1}(s, \tau) \right| \leq K, \tag{68}$$

$$\left| \sum_{j_1, j_2=0}^{p} C_{j_1 j_2 j_2 j_1}(s, \tau) \right| \leq K, \tag{69}$$

$$\left| \sum_{j_1, j_2=0}^{p} C_{j_2 j_1 j_1}(s, \tau) C_{j_2}(\theta, u) \right| \leq K, \tag{70}$$

$$\left| \sum_{j_1, j_2=0}^{p} C_{j_1 j_2 j_1}(s, \tau) C_{j_2}(\theta, u) \right| \leq K, \tag{71}$$

$$\left| \sum_{j_1, j_2=0}^{p} C_{j_2 j_2 j_1}(s, \tau) C_{j_1}(\theta, u) \right| \leq K, \tag{72}$$

$$\left|\sum_{j_1,j_2=0}^{p} C_{j_1j_1}(s,\tau)C_{j_2j_2}(\theta,u)\right| \le K, \tag{73}$$

$$\left|\sum_{j_1,j_2=0}^{p} C_{j_2j_1}(s,\tau)C_{j_2j_1}(\theta,u)\right| \le K, \tag{74}$$

$$\left|\sum_{j_1,j_2=0}^{p} C_{j_2j_1}(s,\tau)C_{j_1j_2}(\theta,u)\right| \le K, \tag{75}$$

$$\left|\sum_{j_1,j_2=0}^{p} C_{j_1}(s,\tau)C_{j_1}(\rho,v)C_{j_2j_2}(\theta,u)\right| \le K, \tag{76}$$

$$\left|\sum_{j_1,j_2=0}^{p} C_{j_1}(s,\tau)C_{j_2}(\rho,v)C_{j_1j_2}(\theta,u)\right| \le K \tag{77}$$

together with the following condition:

$$\left|\sum_{j_1,j_2=0}^{p} C_{j_1}(s,\tau)C_{j_2}(\rho,v)C_{j_1}(\theta,u)C_{j_2}(\mu,w)\right| \le K \tag{78}$$

cover the case $k = 7$, $r = 1, 2$ (see (63)), where $p \in \mathbf{N}$, $t \le \tau < s \le T$, $t \le u < \theta \le T$, $t \le v < \rho \le T$, $t \le w < \mu \le T$, constant K does not depend on $p, s, \tau, u, \theta, v, \rho, w, \mu$ (but only on t, T).

The equalities (65)–(77) are proved in [15] (also see [16], Sect. 2.32) for the case when $\{\phi_j(x)\}_{j=0}^{\infty}$ is an arbitrary CONS in $L_2([t,T])$. The inequality (78) follows from (66).

Now let us focus on the proof of (63) for the case $k = 7$ and $r = 3$. So, we need to prove that

$$\left|\sum_{j_{g_1},j_{g_3},j_{g_5}=0}^{p} C_{j_{d_1}j_{d_1-1}j_{d_1-2}j_{d_1-3}j_{d_1-4}j_{d_1-5}}(s,\tau)\Bigg|_{j_{g_1}=j_{g_2},j_{g_3}=j_{g_4},j_{g_5}=j_{g_6}}\right| \le K, \tag{79}$$

$$\left|\sum_{j_{g_1},j_{g_3},j_{g_5}=0}^{p} \left(C_{j_{d_2}j_{d_2-1}j_{d_2-2}j_{d_2-3}j_{d_2-4}}(s,\tau)C_{j_{d_1}}(\theta,u)\right)\Bigg|_{j_{g_1}=j_{g_2},j_{g_3}=j_{g_4},j_{g_5}=j_{g_6}}\right| \le K, \tag{80}$$

$$\left|\sum_{j_{g_1},j_{g_3},j_{g_5}=0}^{p}\left(C_{j_{d_2}j_{d_2-1}j_{d_2-2}j_{d_2-3}}(s,\tau)C_{j_{d_1}j_{d_1-1}}(\theta,u)\right)\Bigg|_{j_{g_1}=j_{g_2},j_{g_3}=j_{g_4},j_{g_5}=j_{g_6}}\right|\le K, \quad (81)$$

$$\left|\sum_{j_{g_1},j_{g_3},j_{g_5}=0}^{p}\left(C_{j_{d_2}j_{d_2-1}j_{d_2-2}}(s,\tau)C_{j_{d_1}j_{d_1-1}j_{d_1-2}}(\theta,u)\right)\Bigg|_{j_{g_1}=j_{g_2},j_{g_3}=j_{g_4},j_{g_5}=j_{g_6}}\right|\le K, \quad (82)$$

where $p\in\mathbf{N}$, $t\le\tau<s\le T$, $t\le u<\theta\le T$, constant K does not depend on p,s,τ,u,θ (but only on t,T).

The inequalities (80)–(82) are proved using the same technique as inequalities (65)–(77) [15] (also see [16], Sect. 2.32). Here we will only prove as an example the following special case of the inequality (81):

$$\left|\sum_{j_1,j_2,j_3=0}^{p}C_{j_2j_3j_2j_1}(s,\tau)C_{j_3j_1}(\theta,u)\right|\le K. \quad (83)$$

Using the Cauchy–Bunyakovsky inequality as well as Fubini's Theorem, Parseval's equality and (66), we have

$$\left(\sum_{j_1,j_2,j_3=0}^{p}C_{j_2j_3j_2j_1}(s,\tau)C_{j_3j_1}(\theta,u)\right)^2\le$$

$$\le\sum_{j_1,j_3=0}^{p}\left(\sum_{j_2=0}^{p}C_{j_2j_3j_2j_1}(s,\tau)\right)^2\sum_{j_1,j_3=0}^{p}C_{j_3j_1}^2(\theta,u)\le$$

$$\le\sum_{j_1,j_3=0}^{\infty}\left(\sum_{j_2=0}^{p}\int\limits_{\tau}^{s}\phi_{j_2}(u)\int\limits_{\tau}^{u}\phi_{j_3}(z)\int\limits_{\tau}^{z}\phi_{j_2}(y)\int\limits_{\tau}^{y}\phi_{j_1}(x)dxdydzdu\right)^2\times$$

$$\times\sum_{j_1,j_3=0}^{\infty}C_{j_3j_1}^2(\theta,u)=$$

$$=\sum_{j_1,j_3=0}^{\infty}\left(\sum_{j_2=0}^{p}\int\limits_{\tau}^{s}\phi_{j_3}(z)\int\limits_{\tau}^{z}\phi_{j_2}(y)\int\limits_{\tau}^{y}\phi_{j_1}(x)dxdy\int\limits_{z}^{s}\phi_{j_2}(u)dudz\right)^2\cdot\frac{(\theta-u)^2}{2}=$$

$$=\frac{(\theta-u)^2}{2}\sum_{j_1,j_3=0}^{\infty}\left(\sum_{j_2=0}^{p}\int\limits_{\tau}^{s}\phi_{j_3}(z)\int\limits_{\tau}^{z}\phi_{j_1}(x)\int\limits_{x}^{z}\phi_{j_2}(y)dydx\int\limits_{z}^{s}\phi_{j_2}(u)dudz\right)^2=$$

$$= \frac{(\theta-u)^2}{2} \sum_{j_1,j_3=0}^{\infty} \left(\int\limits_{\tau}^{s} \phi_{j_3}(z) \int\limits_{\tau}^{z} \phi_{j_1}(x) \sum_{j_2=0}^{p} C_{j_2}(z,x) C_{j_2}(s,z) dx dz \right)^2 =$$

$$= \frac{(\theta-u)^2}{2} \int\limits_{\tau}^{s} \int\limits_{\tau}^{z} \left(\sum_{j_2=0}^{p} C_{j_2}(z,x) C_{j_2}(s,z) \right)^2 dx dz \le$$

$$\le K^2 \frac{(\theta-u)^2}{2} \frac{(s-\tau)^2}{2} \le K^2 \frac{(T-t)^4}{4} = K_1. \tag{84}$$

The inequality (83) is proved.

The main difficulty is related to the proof of the inequality (79). Further, we prove (79) for all 15 possible cases under the assumption that $\{\phi_j(x)\}_{j=0}^{\infty}$ is a CONS of Legendre polynomials or trigonometric Fourier basis in $L_2([t,T])$. As we noted above, in some situations we will need a function $F \in L_1([t,T])$ (see (64)) instead of a constant K^2 for the polynomial case.

It is easy to see that (79) reduces to the following 15 inequalities:

$$\left| \sum_{j_1,j_2,j_3=0}^{p} C_{j_3 j_2 j_1 j_3 j_2 j_1}(s,\tau) \right| \le K, \tag{85}$$

$$\left| \sum_{j_1,j_2,j_3=0}^{p} C_{j_1 j_3 j_2 j_3 j_2 j_1}(s,\tau) \right| \le K, \tag{86}$$

$$\left| \sum_{j_1,j_2,j_3=0}^{p} C_{j_3 j_2 j_3 j_1 j_2 j_1}(s,\tau) \right| \le K, \tag{87}$$

$$\left| \sum_{j_1,j_2,j_3=0}^{p} C_{j_1 j_2 j_3 j_3 j_2 j_1}(s,\tau) \right| \le K, \tag{88}$$

$$\left| \sum_{j_1,j_2,j_3=0}^{p} C_{j_1 j_2 j_2 j_3 j_3 j_1}(s,\tau) \right| \le K, \tag{89}$$

$$\left| \sum_{j_1,j_2,j_3=0}^{p} C_{j_3 j_3 j_2 j_2 j_1 j_1}(s,\tau) \right| \le K, \tag{90}$$

$$\left|\sum_{j_1,j_2,j_3=0}^{p} C_{j_2j_3j_3j_2j_1j_1}(s,\tau)\right| \le K, \tag{91}$$

$$\left|\sum_{j_1,j_2,j_3=0}^{p} C_{j_3j_2j_3j_2j_1j_1}(s,\tau)\right| \le K, \tag{92}$$

$$\left|\sum_{j_1,j_2,j_3=0}^{p} C_{j_3j_3j_2j_1j_2j_1}(s,\tau)\right| \le K, \tag{93}$$

$$\left|\sum_{j_1,j_2,j_3=0}^{p} C_{j_3j_3j_1j_2j_2j_1}(s,\tau)\right| \le K, \tag{94}$$

$$\left|\sum_{j_1,j_2,j_3=0}^{p} C_{j_2j_1j_3j_3j_2j_1}(s,\tau)\right| \le K, \tag{95}$$

$$\left|\sum_{j_1,j_2,j_3=0}^{p} C_{j_3j_1j_2j_3j_2j_1}(s,\tau)\right| \le K, \tag{96}$$

$$\left|\sum_{j_1,j_2,j_3=0}^{p} C_{j_2j_3j_1j_3j_2j_1}(s,\tau)\right| \le K, \tag{97}$$

$$\left|\sum_{j_1,j_2,j_3=0}^{p} C_{j_3j_1j_3j_2j_2j_1}(s,\tau)\right| \le K, \tag{98}$$

$$\left|\sum_{j_1,j_2,j_3=0}^{p} C_{j_2j_3j_3j_1j_2j_1}(s,\tau)\right| \le K, \tag{99}$$

where $p \in \mathbf{N}$, $t \le \tau < s \le T$, constant K does not depend on p, s, τ (but only on t, T).

Using Fubini's Theorem, we obtain

$$\int_t^T h_6(t_6) \int_t^{t_6} h_5(t_5) \ldots \int_t^{t_2} h_1(t_1) \int_t^{t_1} h_7(t_7) dt_7 dt_1 \ldots dt_6 =$$

$$= \int_t^T h_7(t_7) \int_{t_7}^T h_1(t_1) \int_{t_1}^T h_2(t_2) \ldots \int_{t_5}^T h_6(t_6) dt_6 \ldots dt_1 dt_7 =$$

$$= \int\limits_t^T h_7(t_7) \left(\int\limits_{t_7}^T h_6(t_6) \int\limits_{t_7}^{t_6} h_5(t_5) \ldots \int\limits_{t_7}^{t_2} h_1(t_1) dt_1 \ldots dt_6 \right) dt_7 =$$

$$= \int\limits_t^T h_7(\tau) \left(\int\limits_{\tau}^T h_6(t_6) \int\limits_{\tau}^{t_6} h_5(t_5) \ldots \int\limits_{\tau}^{t_2} h_1(t_1) dt_1 \ldots dt_6 \right) d\tau, \qquad (100)$$

where $h_1(\tau), \ldots, h_7(\tau) \in L_2([t,T])$.

Moreover,

$$\int\limits_t^T h_7(t_7) \int\limits_t^{t_7} h_6(t_6) \ldots \int\limits_t^{t_2} h_1(t_1) dt_1 \ldots dt_6 dt_7 =$$

$$= \int\limits_t^T h_7(s) \left(\int\limits_t^s h_6(t_6) \ldots \int\limits_t^{t_2} h_1(t_1) dt_1 \ldots dt_6 \right) ds. \qquad (101)$$

Taking into account (100) and (101), we note that the conditions (85)–(99) need to be proved in two cases: 1. $\tau = t$, 2. $s = T$. Further, we will not carry out such a refinement if some estimate from (85)–(99) is true for all $\tau, s \in [t,T]$ ($\tau < s$). Looking ahead, we note that consideration of Cases 1 and 2 will be required only for some inequalities from (85)–(99) for the polynomial case.

In [15] (also see [16], Sect. 2.35), the following inequality:

$$\left| \sum_{j_r, j_{r-2}, \ldots, j_2 = 0}^{p} C_{j_r j_r j_{r-2} j_{r-2} \ldots j_2 j_2}(s,\tau) \right| \le K \qquad (102)$$

is proved, where $t \le \tau < s \le T$, $p \in \mathbf{N}$, $r = 2, 4, 6, \ldots$, constant K does not depend on p, s, τ (but only on t, T).

The relation (90) is a particular case of (102). Let us prove (85)–(89), (91)–(99).

Step 1. First, we prove (85)–(89), (95) using special symmetry properties of the Fourier coefficients.

By analogy, with the formula (2.848) (see [16], Sect. 2.14) we have

$$C_{j_6j_5j_4j_3j_2j_1}(s,\tau)+C_{j_1j_2j_3j_4j_5j_6}(s,\tau)=$$
$$=C_{j_6}(s,\tau)C_{j_5j_4j_3j_2j_1}(s,\tau)-C_{j_5j_6}(s,\tau)C_{j_4j_3j_2j_1}(s,\tau)+$$
$$+C_{j_4j_5j_6}(s,\tau)C_{j_3j_2j_1}(s,\tau)-C_{j_3j_4j_5j_6}(s,\tau)C_{j_2j_1}(s,\tau)+$$
$$+C_{j_2j_3j_4j_5j_6}(s,\tau)C_{j_1}(s,\tau). \tag{103}$$

Substituting $j_1=j_4, j_2=j_5, j_3=j_6$ into (103) and summing over j_1,j_2,j_3, we have

$$\sum_{j_1,j_2,j_3=0}^{p} C_{j_3j_2j_1j_3j_2j_1}(s,\tau)+\sum_{j_1,j_2,j_3=0}^{p} C_{j_1j_2j_3j_1j_2j_3}(s,\tau)=2\sum_{j_1,j_2,j_3=0}^{p} C_{j_3j_2j_1j_3j_2j_1}(s,\tau)=$$
$$=\sum_{j_1,j_2,j_3=0}^{p}\Bigg(C_{j_3}(s,\tau)C_{j_2j_1j_3j_2j_1}(s,\tau)-C_{j_2j_3}(s,\tau)C_{j_1j_3j_2j_1}(s,\tau)+$$
$$+C_{j_1j_2j_3}(s,\tau)C_{j_3j_2j_1}(s,\tau)-C_{j_3j_1j_2j_3}(s,\tau)C_{j_2j_1}(s,\tau)+C_{j_2j_3j_1j_2j_3}(s,\tau)C_{j_1}(s,\tau)\Bigg). \tag{104}$$

Then

$$\sum_{j_1,j_2,j_3=0}^{p} C_{j_3j_2j_1j_3j_2j_1}(s,\tau)=\frac{1}{2}\sum_{j_1,j_2,j_3=0}^{p}\Bigg(C_{j_3}(s,\tau)C_{j_2j_1j_3j_2j_1}(s,\tau)-$$
$$-C_{j_2j_3}(s,\tau)C_{j_1j_3j_2j_1}(s,\tau)+C_{j_1j_2j_3}(s,\tau)C_{j_3j_2j_1}(s,\tau)-$$
$$-C_{j_3j_1j_2j_3}(s,\tau)C_{j_2j_1}(s,\tau)+C_{j_2j_3j_1j_2j_3}(s,\tau)C_{j_1}(s,\tau)\Bigg). \tag{105}$$

Similarly, we get

$$\sum_{j_1,j_2,j_3=0}^{p} C_{j_1j_3j_2j_3j_2j_1}(s,\tau)=\frac{1}{2}\sum_{j_1,j_2,j_3=0}^{p}\Bigg(C_{j_1}(s,\tau)C_{j_3j_2j_3j_2j_1}(s,\tau)-$$
$$-C_{j_3j_1}(s,\tau)C_{j_2j_3j_2j_1}(s,\tau)+C_{j_2j_3j_1}(s,\tau)C_{j_3j_2j_1}(s,\tau)-$$
$$-C_{j_3j_2j_3j_1}(s,\tau)C_{j_2j_1}(s,\tau)+C_{j_2j_3j_2j_3j_1}(s,\tau)C_{j_1}(s,\tau)\Bigg), \tag{106}$$

$$\sum_{j_1,j_2,j_3=0}^{p} C_{j_3j_2j_3j_1j_2j_1}(s,\tau) = \frac{1}{2}\sum_{j_1,j_2,j_3=0}^{p}\Big(C_{j_3}(s,\tau)C_{j_2j_3j_1j_2j_1}(s,\tau)-$$
$$-C_{j_2j_3}(s,\tau)C_{j_3j_1j_2j_1}(s,\tau)+C_{j_3j_2j_3}(s,\tau)C_{j_1j_2j_1}(s,\tau)-$$
$$-C_{j_1j_3j_2j_3}(s,\tau)C_{j_2j_1}(s,\tau)+C_{j_2j_1j_3j_2j_3}(s,\tau)C_{j_1}(s,\tau)\Big), \tag{107}$$

$$\sum_{j_1,j_2,j_3=0}^{p} C_{j_1j_2j_3j_3j_2j_1}(s,\tau) = \frac{1}{2}\sum_{j_1,j_2,j_3=0}^{p}\Big(C_{j_1}(s,\tau)C_{j_2j_3j_3j_2j_1}(s,\tau)-$$
$$-C_{j_2j_1}(s,\tau)C_{j_3j_3j_2j_1}(s,\tau)+(C_{j_3j_2j_1}(s,\tau))^2-$$
$$-C_{j_3j_3j_2j_1}(s,\tau)C_{j_2j_1}(s,\tau)+C_{j_2j_3j_3j_2j_1}(s,\tau)C_{j_1}(s,\tau)\Big), \tag{108}$$

$$\sum_{j_1,j_2,j_3=0}^{p} C_{j_1j_3j_3j_2j_2j_1}(s,\tau) = \frac{1}{2}\sum_{j_1,j_2,j_3=0}^{p}\Big(C_{j_1}(s,\tau)C_{j_3j_3j_2j_2j_1}(s,\tau)-$$
$$-C_{j_3j_1}(s,\tau)C_{j_3j_2j_2j_1}(s,\tau)+C_{j_3j_3j_1}(s,\tau)C_{j_2j_2j_1}(s,\tau)-$$
$$-C_{j_2j_3j_3j_1}(s,\tau)C_{j_2j_1}(s,\tau)+C_{j_2j_2j_3j_3j_1}(s,\tau)C_{j_1}(s,\tau)\Big), \tag{109}$$

$$\sum_{j_1,j_2,j_3=0}^{p} C_{j_2j_1j_3j_3j_2j_1}(s,\tau) = \frac{1}{2}\sum_{j_1,j_2,j_3=0}^{p}\Big(C_{j_2}(s,\tau)C_{j_1j_3j_3j_2j_1}(s,\tau)-$$
$$-C_{j_1j_2}(s,\tau)C_{j_3j_3j_2j_1}(s,\tau)+C_{j_3j_1j_2}(s,\tau)C_{j_3j_2j_1}(s,\tau)-$$
$$-C_{j_3j_3j_1j_2}(s,\tau)C_{j_2j_1}(s,\tau)+C_{j_2j_3j_3j_1j_2}(s,\tau)C_{j_1}(s,\tau)\Big). \tag{110}$$

Using to the right-hand sides of (105)–(110) the technique that led to the estimate (84), we obtain the inequalities (85)–(89), (95).

Step 2. It is not difficult to see that

$$\sum_{j_1,j_2,j_3=0}^{p} C_{j_3j_3j_1j_2j_2j_1}(s,\tau) = \sum_{j_1,j_2,j_3=0}^{p} C_{j_1j_1j_2j_3j_3j_2}(s,\tau), \tag{111}$$

$$\sum_{j_1,j_2,j_3=0}^{p} C_{j_3j_3j_2j_1j_2j_1}(s,\tau) = \sum_{j_1,j_2,j_3=0}^{p} C_{j_1j_1j_2j_3j_2j_3}(s,\tau), \tag{112}$$

$$\sum_{j_1,j_2,j_3=0}^{p} C_{j_2j_3j_3j_1j_2j_1}(s,\tau) = \sum_{j_1,j_2,j_3=0}^{p} C_{j_1j_2j_2j_3j_1j_3}(s,\tau). \tag{113}$$

Further, using (111)–(113) and (103), we get

$$\sum_{j_1,j_2,j_3=0}^{p} C_{j_2j_3j_3j_2j_1j_1}(s,\tau) + \sum_{j_1,j_2,j_3=0}^{p} C_{j_3j_3j_1j_2j_2j_1}(s,\tau) =$$

$$= \sum_{j_1,j_2,j_3=0}^{p} C_{j_2j_3j_3j_2j_1j_1}(s,\tau) + \sum_{j_1,j_2,j_3=0}^{p} C_{j_1j_1j_2j_3j_3j_2}(s,\tau) =$$

$$= \sum_{j_1,j_2,j_3=0}^{p} \Bigg(C_{j_2}(s,\tau) C_{j_3j_3j_2j_1j_1}(s,\tau) -$$

$$- C_{j_3j_2}(s,\tau) C_{j_3j_2j_1j_1}(s,\tau) + C_{j_3j_3j_2}(s,\tau) C_{j_2j_1j_1}(s,\tau) -$$

$$- C_{j_2j_3j_3j_2}(s,\tau) C_{j_1j_1}(s,\tau) + C_{j_1j_2j_3j_3j_2}(s,\tau) C_{j_1}(s,\tau) \Bigg), \tag{114}$$

$$\sum_{j_1,j_2,j_3=0}^{p} C_{j_3j_2j_3j_2j_1j_1}(s,\tau) + \sum_{j_1,j_2,j_3=0}^{p} C_{j_3j_3j_2j_1j_2j_1}(s,\tau) =$$

$$= \sum_{j_1,j_2,j_3=0}^{p} C_{j_3j_2j_3j_2j_1j_1}(s,\tau) + \sum_{j_1,j_2,j_3=0}^{p} C_{j_1j_1j_2j_3j_2j_3}(s,\tau) =$$

$$= \sum_{j_1,j_2,j_3=0}^{p} \Bigg(C_{j_3}(s,\tau) C_{j_2j_3j_2j_1j_1}(s,\tau) -$$

$$- C_{j_2j_3}(s,\tau) C_{j_3j_2j_1j_1}(s,\tau) + C_{j_3j_2j_3}(s,\tau) C_{j_2j_1j_1}(s,\tau) -$$

$$- C_{j_2j_3j_2j_3}(s,\tau) C_{j_1j_1}(s,\tau) + C_{j_1j_2j_3j_2j_3}(s,\tau) C_{j_1}(s,\tau) \Bigg), \tag{115}$$

$$\sum_{j_1,j_2,j_3=0}^{p} C_{j_3j_1j_3j_2j_2j_1}(s,\tau) + \sum_{j_1,j_2,j_3=0}^{p} C_{j_2j_3j_3j_1j_2j_1}(s,\tau) =$$

$$= \sum_{j_1,j_2,j_3=0}^{p} C_{j_3j_1j_3j_2j_2j_1}(s,\tau) + \sum_{j_1,j_2,j_3=0}^{p} C_{j_1j_2j_2j_3j_1j_3}(s,\tau) =$$

$$= \sum_{j_1,j_2,j_3=0}^{p} \Biggl(C_{j_3}(s,\tau) C_{j_1j_3j_2j_2j_1}(s,\tau) -$$

$$- C_{j_1j_3}(s,\tau) C_{j_3j_2j_2j_1}(s,\tau) + C_{j_3j_1j_3}(s,\tau) C_{j_2j_2j_1}(s,\tau) -$$

$$- C_{j_2j_3j_1j_3}(s,\tau) C_{j_2j_1}(s,\tau) + C_{j_2j_2j_3j_1j_3}(s,\tau) C_{j_1}(s,\tau) \Biggr). \tag{116}$$

Applying to the right-hand sides of (114)–(116) the technique that led to the estimate (84), we obtain the inequalities

$$\left| \sum_{j_1,j_2,j_3=0}^{p} C_{j_2j_3j_3j_2j_1j_1}(s,\tau) + \sum_{j_1,j_2,j_3=0}^{p} C_{j_3j_3j_1j_2j_2j_1}(s,\tau) \right| \le K, \tag{117}$$

$$\left| \sum_{j_1,j_2,j_3=0}^{p} C_{j_3j_2j_3j_2j_1j_1}(s,\tau) + \sum_{j_1,j_2,j_3=0}^{p} C_{j_3j_3j_2j_1j_2j_1}(s,\tau) \right| \le K, \tag{118}$$

$$\left| \sum_{j_1,j_2,j_3=0}^{p} C_{j_3j_1j_3j_2j_2j_1}(s,\tau) + \sum_{j_1,j_2,j_3=0}^{p} C_{j_2j_3j_3j_1j_2j_1}(s,\tau) \right| \le K, \tag{119}$$

where $p \in \mathbf{N}$, $t \le \tau < s \le T$, constant K does not depend on p, s, τ (but only on t, T) and may differ from line to line.

Note that $|a| \le K_1 + K$ follows from $|b| \le K$ and $|a+b| \le K_1$, where $a, b, K, K_1 \in \mathbf{R}$. Indeed, we have $|a| = |a+b-b| \le |a+b| + |b| \le K_1 + K$. Then from (117)–(119) it follows that if we prove (93), (94), (99), then (92), (91), (98) will be proved. Thus, it remains to prove (93), (94), (96), (97), (99).

Step 3. Let us prove (93), (94), (96), (97), (99). Consider (97). Recall the equality (see the proof in [16] (Sect. 2.10, 2.18) or [11])

$$\sum_{j_l=0}^{\infty} C_{j_k\ldots j_{l+1}j_lj_{l-1}\ldots j_{s+1}j_lj_{s-1}\ldots j_1} = 0 \tag{120}$$

or

$$\sum_{j_l=0}^{p} C_{j_k\ldots j_{l+1}j_lj_{l-1}\ldots j_{s+1}j_lj_{s-1}\ldots j_1} = -\sum_{j_l=p+1}^{\infty} C_{j_k\ldots j_{l+1}j_lj_{l-1}\ldots j_{s+1}j_lj_{s-1}\ldots j_1}, \tag{121}$$

where $C_{j_k\ldots j_1}$ is defined by (2) and $l-1\ge s+1$.

Using the Cauchy–Bunyakovsky inequality as well as Fubini's Theorem, Parseval's equality, (121), (66), and Lebesgue's Dominated Convergence Theorem, we have

$$\left(\sum_{j_1,j_2,j_3=0}^{p} C_{j_2j_3j_1j_3j_2j_1}(s,\tau)\right)^2=\left(\sum_{j_2=0}^{p}1\cdot\sum_{j_1,j_3=0}^{p} C_{j_2j_3j_1j_3j_2j_1}(s,\tau)\right)^2\le$$

$$\le\sum_{j_2=0}^{p}1^2\cdot\sum_{j_2=0}^{p}\left(\sum_{j_1,j_3=0}^{p} C_{j_2j_3j_1j_3j_2j_1}(s,\tau)\right)^2=$$

$$=(p+1)\sum_{j_2=0}^{p}\left(\sum_{j_1,j_3=0}^{p} C_{j_2j_3j_1j_3j_2j_1}(s,\tau)\right)^2=$$

$$=(p+1)\sum_{j_2=0}^{p}\left(\sum_{j_1,j_3=0}^{p}\int\limits_{\tau}^{s}\phi_{j_2}(t_6)\int\limits_{\tau}^{t_6}\phi_{j_2}(t_2)C_{j_1}(t_2,\tau)C_{j_3j_1j_3}(t_6,t_2)dt_2dt_6\right)^2\le$$

$$\le(p+1)\sum_{j_2,j_2'=0}^{p}\left(\sum_{j_1,j_3=0}^{p}\int\limits_{\tau}^{s}\phi_{j_2}(t_6)\int\limits_{\tau}^{t_6}\phi_{j_2'}(t_2)C_{j_1}(t_2,\tau)C_{j_3j_1j_3}(t_6,t_2)dt_2dt_6\right)^2\le$$

$$\le(p+1)\sum_{j_2,j_2'=0}^{\infty}\left(\int\limits_{\tau}^{s}\phi_{j_2}(t_6)\int\limits_{\tau}^{t_6}\phi_{j_2'}(t_2)\sum_{j_1,j_3=0}^{p}C_{j_1}(t_2,\tau)C_{j_3j_1j_3}(t_6,t_2)dt_2dt_6\right)^2=$$

$$=(p+1)\int\limits_{\tau}^{s}\int\limits_{\tau}^{t_6}\left(\sum_{j_1=0}^{p}C_{j_1}(t_2,\tau)\sum_{j_3=0}^{p}C_{j_3j_1j_3}(t_6,t_2)\right)^2dt_2dt_6=$$

$$=(p+1)\int\limits_{\tau}^{s}\int\limits_{\tau}^{t_6}\left(\sum_{j_1=0}^{p}C_{j_1}(t_2,\tau)\sum_{j_3=p+1}^{\infty}C_{j_3j_1j_3}(t_6,t_2)\right)^2dt_2dt_6\le$$

$$\le(p+1)\int\limits_{\tau}^{s}\int\limits_{\tau}^{t_6}\sum_{j_1=0}^{p}C_{j_1}^2(t_2,\tau)\sum_{j_1=0}^{p}\left(\sum_{j_3=p+1}^{\infty}C_{j_3j_1j_3}(t_6,t_2)\right)^2dt_2dt_6\le$$

$$\le(p+1)\int\limits_{\tau}^{s}\int\limits_{\tau}^{t_6}\sum_{j_1=0}^{\infty}C_{j_1}^2(t_2,\tau)\sum_{j_1=0}^{p}\left(\sum_{j_3=p+1}^{\infty}C_{j_3j_1j_3}(t_6,t_2)\right)^2dt_2dt_6=$$

$$\leq (p+1)\int\limits_{\tau}^{s}\int\limits_{\tau}^{t_6}(t_2-\tau)\sum_{j_1=0}^{p}\left(\sum_{j_3=p+1}^{\infty}C_{j_3j_1j_3}(t_6,t_2)\right)^2 dt_2dt_6 =$$

$$= (p+1)\int\limits_{\tau}^{s}\int\limits_{\tau}^{t_6}(t_2-\tau)\sum_{j_1=0}^{p}\left(\sum_{j_3=p+1}^{\infty}\int\limits_{t_2}^{t_6}\phi_{j_1}(\theta)C_{j_3}(\theta,t_2)C_{j_3}(t_6,\theta)d\theta\right)^2 dt_2dt_6 =$$

$$= (p+1)\int\limits_{\tau}^{s}\int\limits_{\tau}^{t_6}(t_2-\tau)\sum_{j_1=0}^{p}\left(\int\limits_{t_2}^{t_6}\phi_{j_1}(\theta)\sum_{j_3=p+1}^{\infty}C_{j_3}(\theta,t_2)C_{j_3}(t_6,\theta)d\theta\right)^2 dt_2dt_6 \leq$$

$$\leq (p+1)\int\limits_{\tau}^{s}\int\limits_{\tau}^{t_6}(t_2-\tau)\sum_{j_1=0}^{\infty}\left(\int\limits_{t_2}^{t_6}\phi_{j_1}(\theta)\sum_{j_3=p+1}^{\infty}C_{j_3}(\theta,t_2)C_{j_3}(t_6,\theta)d\theta\right)^2 dt_2dt_6 =$$

$$= (p+1)\int\limits_{\tau}^{s}\int\limits_{\tau}^{t_6}(t_2-\tau)\int\limits_{t_2}^{t_6}\left(\sum_{j_3=p+1}^{\infty}C_{j_3}(\theta,t_2)C_{j_3}(t_6,\theta)\right)^2 d\theta dt_2dt_6. \quad (122)$$

Consider the trigonometric Fourier basis in $L_2([t,T])$

$$\phi_j(\tau)=\frac{1}{\sqrt{T-t}}\begin{cases}1, & j=0\\ \sqrt{2}\sin\left(2\pi r(\tau-t)/(T-t)\right), & j=2r-1,\\ \sqrt{2}\cos\left(2\pi r(\tau-t)/(T-t)\right), & j=2r\end{cases}$$

where $r=1,2,\ldots$

For the trigonometric case, we have the following obvious estimate:

$$|C_j(x,v)|=\left|\int\limits_{v}^{x}\phi_j(\tau)d\tau\right|<\frac{C}{j}\quad (j>0), \quad (123)$$

where constant C does not depend on j,x,v.

Note that

$$\sum_{j=p+1}^{\infty}\frac{1}{j^2}\leq\int\limits_{p}^{\infty}\frac{dx}{x^2}=\frac{1}{p}. \quad (124)$$

Combining (122)–(124), we get

$$\left(\sum_{j_1,j_2,j_3=0}^{p} C_{j_2j_3j_1j_3j_2j_1}(s,\tau)\right)^2 \le \frac{K_1(p+1)}{p^2} \le K^2,$$

where constants K, K_1 depend only on t, T. The inequality (97) is proved for the trigonometric case.

Consider the CONS of Legendre polynomials in $L_2([t,T])$

$$\phi_j(\tau) = \sqrt{\frac{2j+1}{T-t}} P_j\left(\left(\tau - \frac{T+t}{2}\right)\frac{2}{T-t}\right),$$

where

$$P_j(y) = \frac{1}{2^j j!}\frac{d^j}{dy^j}\left(y^2-1\right)^j \quad (j=0,1,2,\ldots)$$

is the Legendre polynomial.

For the polynomial case, we have the following estimate (see [16], Sect. 2.36):

$$|C_j(x,v)| = \left|\int_v^x \phi_j(\tau)d\tau\right| < \frac{C}{j^{1-\varepsilon/2}}\left(\frac{1}{(1-z^2(x))^{1/4-\varepsilon/4}} + \frac{1}{(1-z^2(v))^{1/4-\varepsilon/4}}\right), \tag{125}$$

where $j \in \mathbf{N}$, $z(x), z(v) \in (-1,1)$,

$$z(x) = \left(x - \frac{T+t}{2}\right)\frac{2}{T-t}, \tag{126}$$

$x, v \in (t,T)$, $\varepsilon \in (0,1)$ is an arbitrary small positive real number, constant C does not depend on j.

Note that

$$\sum_{j=p+1}^{\infty}\frac{1}{j^{2-\varepsilon}} \le \int_p^{\infty}\frac{dx}{x^{2-\varepsilon}} = \frac{1}{(1-\varepsilon)p^{1-\varepsilon}}. \tag{127}$$

Combining (122), (125), (127) ($\varepsilon = 1/4$), we obtain

$$\left(\sum_{j_1,j_2,j_3=0}^{p} C_{j_2j_3j_1j_3j_2j_1}(s,\tau)\right)^2 \le \frac{K_1(p+1)}{p^{3/2}} \le K^2,$$

where constants K, K_1 depend only on t, T. The inequality (97) is proved for the polynomial case.

Let us prove (96). In complete analogy with the proof of (97) we have

$$\left(\sum_{j_1,j_2,j_3=0}^{p} C_{j_3j_1j_2j_3j_2j_1}(s,\tau)\right)^2 \leq$$

$$\leq (p+1)\int\limits_{\tau}^{s}(s-t_5)\int\limits_{\tau}^{t_5}\int\limits_{t_1}^{t_5}\left(\sum_{j_2=p+1}^{\infty} C_{j_2}(\theta,t_1)C_{j_2}(t_5,\theta)\right)^2 d\theta dt_1 dt_5.$$

The further proof is the same as in the case of (97). The inequality (96) is proved.

Let us prove (99). By analogy, with the proof of (97) (see (122)) we get

$$\left(\sum_{j_1,j_2,j_3=0}^{p} C_{j_2j_3j_3j_1j_2j_1}(s,\tau)\right)^2 \leq$$

$$\leq (p+1)\int\limits_{\tau}^{s}(s-t_5)\int\limits_{\tau}^{t_5}\int\limits_{\tau}^{t_4}\left(\sum_{j_1=p+1}^{\infty} C_{j_1}(\theta,\tau)C_{j_1}(t_4,\theta)\right)^2 d\theta dt_4 dt_5. \quad (128)$$

The further proof for the trigonometric case is the same as for the inequality (97).

Consider the polynomial case. In this case, we note that it is actually necessary to consider the following two cases of (128) (see (100) and (101)):

$$1.\ \tau = t, \quad 2.\ s = T \quad (\tau = t \text{ or } \tau > t). \quad (129)$$

For Case 1, the estimate (125) is simplified as follows (see [16], Sect. 2.36)

$$|C_j(x,t)| = \left|\int\limits_{t}^{x}\phi_j(\tau)d\tau\right| < \frac{C}{j^{1-\varepsilon/2}}\frac{1}{(1-z^2(x))^{1/4-\varepsilon/4}}, \quad (130)$$

where notations are the same as in (125).

Combining (128), (125), (127), (130) ($\varepsilon = 1/4$), we obtain

$$\left(\sum_{j_1,j_2,j_3=0}^{p} C_{j_2j_3j_3j_1j_2j_1}(s,t)\right)^2 \le \frac{K_1(p+1)}{p^{3/2}} \le K^2, \tag{131}$$

where constants K, K_1 depend only on t, T. The inequality (99) is proved for the polynomial case ($\tau = t$).

Consider Case 2. Combining (128), (125), (127), (130) ($\varepsilon = 1/4$), we obtain

$$\left(\sum_{j_1,j_2,j_3=0}^{p} C_{j_2j_3j_3j_1j_2j_1}(T,\tau)\right)^2 \Bigg|_{\tau>t} \le \frac{K_1(p+1)}{p^{3/2}} \frac{1}{(1-z^2(\tau))^{3/8}} \le$$

$$\le \frac{K^2}{(1-z^2(\tau))^{3/8}} \stackrel{\text{def}}{=} F(\tau), \tag{132}$$

$$\left(\sum_{j_1,j_2,j_3=0}^{p} C_{j_2j_3j_3j_1j_2j_1}(T,\tau)\right)^2 \Bigg|_{\tau=t} \le \frac{K_1(p+1)}{p^{3/2}}, \tag{133}$$

where constants K, K_1 depend only on t, T and $F(\tau) \in L_1([t,T])$ (integrable majorant (see above in this section)).

Combining (132) and (133), we get the following weakened version of the inequality (99):

$$\left(\sum_{j_1,j_2,j_3=0}^{p} C_{j_2j_3j_3j_1j_2j_1}(T,\tau)\right)^2 \le F(\tau) \tag{134}$$

for the polynomial case ($s = T$), where

$$F(\tau) = \frac{K^2}{(1-z^2(\tau))^{3/8}}.$$

Let us prove (94). Using the Cauchy–Bunyakovsky inequality as well as Fubini's Theorem and Parseval's equality, we have

$$\left(\sum_{j_1,j_2,j_3=0}^{p} C_{j_3j_3j_1j_2j_2j_1}(s,\tau)\right)^2 = \left(\sum_{j_3=0}^{p} 1 \cdot \sum_{j_1,j_2=0}^{p} C_{j_3j_3j_1j_2j_2j_1}(s,\tau)\right)^2 \le$$

$$\leq \sum_{j_3=0}^{p} 1^2 \cdot \sum_{j_3=0}^{p} \left(\sum_{j_1,j_2=0}^{p} C_{j_3 j_3 j_1 j_2 j_2 j_1}(s,\tau) \right)^2 =$$

$$= (p+1) \sum_{j_3=0}^{p} \left(\sum_{j_1,j_2=0}^{p} C_{j_3 j_3 j_1 j_2 j_2 j_1}(s,\tau) \right)^2 =$$

$$= (p+1) \sum_{j_3=0}^{p} \left(\sum_{j_1,j_2=0}^{p} \int\limits_{\tau}^{s} \phi_{j_3}(t_6) \int\limits_{\tau}^{t_6} \phi_{j_3}(t_5) C_{j_1 j_2 j_2 j_1}(t_5,\tau) dt_5 dt_6 \right)^2 \leq$$

$$\leq (p+1) \sum_{j_3,j_3'=0}^{p} \left(\int\limits_{\tau}^{s} \phi_{j_3}(t_6) \int\limits_{\tau}^{t_6} \phi_{j_3'}(t_5) \sum_{j_1,j_2=0}^{p} C_{j_1 j_2 j_2 j_1}(t_5,\tau) dt_5 dt_6 \right)^2 \leq$$

$$\leq (p+1) \sum_{j_3,j_3'=0}^{\infty} \left(\int\limits_{\tau}^{s} \phi_{j_3}(t_6) \int\limits_{\tau}^{t_6} \phi_{j_3'}(t_5) \sum_{j_1,j_2=0}^{p} C_{j_1 j_2 j_2 j_1}(t_5,\tau) dt_5 dt_6 \right)^2 =$$

$$= (p+1) \int\limits_{\tau}^{s} \int\limits_{\tau}^{t_6} \left(\sum_{j_1,j_2=0}^{p} C_{j_1 j_2 j_2 j_1}(t_5,\tau) \right)^2 dt_5 dt_6 =$$

$$= (p+1) \int\limits_{\tau}^{s} \int\limits_{\tau}^{t_6} \left(\sum_{j_2=0}^{p} 1 \cdot \sum_{j_1=0}^{p} C_{j_1 j_2 j_2 j_1}(t_5,\tau) \right)^2 dt_5 dt_6 \leq$$

$$\leq (p+1)^2 \int\limits_{\tau}^{s} \int\limits_{\tau}^{t_6} \sum_{j_2=0}^{p} \left(\sum_{j_1=0}^{p} C_{j_1 j_2 j_2 j_1}(t_5,\tau) \right)^2 dt_5 dt_6 =$$

$$= (p+1)^2 \int\limits_{\tau}^{s} \int\limits_{\tau}^{t_6} \sum_{j_2=0}^{p} \left(\sum_{j_1=0}^{p} \int\limits_{\tau}^{t_5} \phi_{j_2}(t_3) \int\limits_{\tau}^{t_3} \phi_{j_2}(t_2) C_{j_1}(t_2,\tau) C_{j_1}(t_5,t_3) dt_2 dt_3 \right)^2 dt_5 dt_6 \leq$$

$$\leq (p+1)^2 \int\limits_{\tau}^{s} \int\limits_{\tau}^{t_6} \sum_{j_2,j_2'=0}^{p} \left(\int\limits_{\tau}^{t_5} \phi_{j_2}(t_3) \times \right.$$

$$\left. \times \int\limits_{\tau}^{t_3} \phi_{j_2'}(t_2) \sum_{j_1=0}^{p} C_{j_1}(t_2,\tau) C_{j_1}(t_5,t_3) dt_2 dt_3 \right)^2 dt_5 dt_6 \leq$$

$$\leq (p+1)^2 \int\limits_{\tau}^{s} \int\limits_{\tau}^{t_6} \sum_{j_2, j_2'=0}^{\infty} \left(\int\limits_{\tau}^{t_5} \phi_{j_2}(t_3) \times \right.$$

$$\left. \times \int\limits_{\tau}^{t_3} \phi_{j_2'}(t_2) \left(\sum_{j_1=0}^{\infty} - \sum_{j_1=p+1}^{\infty} \right) C_{j_1}(t_2, \tau) C_{j_1}(t_5, t_3) dt_2 dt_3 \right)^2 dt_5 dt_6 =$$

$$= (p+1)^2 \int\limits_{\tau}^{s} \int\limits_{\tau}^{t_6} \int\limits_{\tau}^{t_5} \int\limits_{\tau}^{t_3} \left(\sum_{j_1=p+1}^{\infty} C_{j_1}(t_2, \tau) C_{j_1}(t_5, t_3) \right)^2 dt_2 dt_3 dt_5 dt_6. \quad (135)$$

Consider the trigonometric case. Combining (135), (123), (124), we obtain

$$\left(\sum_{j_1, j_2, j_3=0}^{p} C_{j_3 j_3 j_1 j_2 j_2 j_1}(s, \tau) \right)^2 \leq \frac{K_1 (p+1)^2}{p^2} \leq K^2,$$

where constants K, K_1 depend only on t, T. The inequality (94) is proved for the trigonometric case.

Consider the polynomial case for two cases (129). Let $\tau = t$. The modification of the estimate (125) for $\varepsilon = 0$ is as follows (see [16], Sect. 2.36)

$$|C_j(x, v)| = \left| \int\limits_{v}^{x} \phi_j(\tau) d\tau \right| < \frac{C}{j} \left(\frac{1}{(1 - z^2(x))^{1/4}} + \frac{1}{(1 - z^2(v))^{1/4}} \right), \quad (136)$$

where $j \in \mathbf{N}$, $z(x), z(v) \in (-1, 1)$ ($z(x)$ is defined by (126)), $x, v \in (t, T)$, constant C does not depend on j.

For $v = t$, the estimate (136) is simplified as follows (see [16], Sect. 2.36)

$$|C_j(x, t)| = \left| \int\limits_{t}^{x} \phi_j(\tau) d\tau \right| < \frac{C}{j(1 - z^2(x))^{1/4}}, \quad (137)$$

where notations are the same as in (136).

Combining (135), (136), (137), we get

$$\left(\sum_{j_1, j_2, j_3=0}^{p} C_{j_3 j_3 j_1 j_2 j_2 j_1}(s, t) \right)^2 \leq \frac{K_1 (p+1)^2}{p^2} \leq K^2,$$

where constants K, K_1 depend only on t, T. The inequality (94) is proved for the polynomial case ($\tau = t$).

Now let $s = T$. Using (135), (136), (137), we obtain

$$\left(\sum_{j_1,j_2,j_3=0}^{p} C_{j_3j_3j_1j_2j_2j_1}(T,\tau)\right)^2\Bigg|_{\tau>t} \leq \frac{K_1(p+1)^2}{p^2}\frac{1}{(1-z^2(\tau))^{1/2}} \leq$$

$$\leq \frac{K^2}{(1-z^2(\tau))^{1/2}} \stackrel{\text{def}}{=} F(\tau), \tag{138}$$

$$\left(\sum_{j_1,j_2,j_3=0}^{p} C_{j_3j_3j_1j_2j_2j_1}(T,\tau)\right)^2\Bigg|_{\tau=t} \leq \frac{K_1(p+1)^2}{p^2}, \tag{139}$$

where constants K, K_1 depend only on t, T and $F(\tau) \in L_1([t,T])$ (integrable majorant (see above in this section)).

Combining (138) and (139), we get the following weakened version of the inequality (94):

$$\left(\sum_{j_1,j_2,j_3=0}^{p} C_{j_3j_3j_1j_2j_2j_1}(T,\tau)\right)^2 \leq F(\tau) \tag{140}$$

for the polynomial case ($s = T$), where

$$F(\tau) = \frac{K^2}{(1-z^2(\tau))^{1/2}}.$$

Finally, we prove the inequality (93). By analogy, with (135) we get

$$\left(\sum_{j_1,j_2,j_3=0}^{p} C_{j_3j_3j_2j_1j_2j_1}(s,\tau)\right)^2 \leq$$

$$\leq (p+1)\sum_{j_3=0}^{p}\left(\sum_{j_1,j_2=0}^{p} C_{j_3j_3j_2j_1j_2j_1}(s,\tau)\right)^2 =$$

$$= (p+1)\sum_{j_3=0}^{p}\left(\sum_{j_1,j_2=0}^{p}\int_{\tau}^{s}\phi_{j_3}(t_6)\int_{\tau}^{t_6}\phi_{j_3}(t_5)C_{j_2j_1j_2j_1}(t_5,\tau)dt_5dt_6\right)^2 \leq$$

$$\le (p+1)\sum_{j_3,j_3'=0}^{\infty}\left(\int\limits_{\tau}^{s}\phi_{j_3}(t_6)\int\limits_{\tau}^{t_6}\phi_{j_3'}(t_5)\sum_{j_1,j_2=0}^{p}C_{j_2j_1j_2j_1}(t_5,\tau)dt_5dt_6\right)^2=$$

$$=(p+1)\int\limits_{\tau}^{s}\int\limits_{\tau}^{t_6}\left(\sum_{j_1,j_2=0}^{p}C_{j_2j_1j_2j_1}(t_5,\tau)\right)^2dt_5dt_6=$$

$$\le (p+1)^2\int\limits_{\tau}^{s}\int\limits_{\tau}^{t_6}\sum_{j_2=0}^{p}\left(\sum_{j_1=0}^{p}C_{j_2j_1j_2j_1}(t_5,\tau)\right)^2dt_5dt_6=$$

$$=(p+1)^2\int\limits_{\tau}^{s}\int\limits_{\tau}^{t_6}\sum_{j_2=0}^{p}\left(\sum_{j_1=0}^{p}\int\limits_{\tau}^{t_5}\phi_{j_2}(t_4)\int\limits_{\tau}^{t_4}\phi_{j_2}(t_2)C_{j_1}(t_2,\tau)C_{j_1}(t_4,t_2)dt_2dt_4\right)^2dt_5dt_6\le$$

$$\le (p+1)^2\int\limits_{\tau}^{s}\int\limits_{\tau}^{t_6}\sum_{j_2,j_2'=0}^{\infty}\left(\int\limits_{\tau}^{t_5}\phi_{j_2}(t_4)\times\right.$$

$$\left.\times\int\limits_{\tau}^{t_4}\phi_{j_2'}(t_2)\left(\sum_{j_1=0}^{\infty}-\sum_{j_1=p+1}^{\infty}\right)C_{j_1}(t_2,\tau)C_{j_1}(t_4,t_2)dt_2dt_4\right)^2dt_5dt_6=$$

$$=(p+1)^2\int\limits_{\tau}^{s}\int\limits_{\tau}^{t_6}\int\limits_{\tau}^{t_5}\int\limits_{\tau}^{t_4}\left(\sum_{j_1=p+1}^{\infty}C_{j_1}(t_2,\tau)C_{j_1}(t_4,t_2)\right)^2dt_2dt_4dt_5dt_6. \quad (141)$$

The further proof of inequality (93) for the trigonometric case and the weakened analogue of inequality (93) for the polynomial case is completely analogous to the proof of (99) and its weakened analogue (see (128), (131), (134)). Thus, Theorem 9 is proved.

3.4 Expansion of Iterated Stratonovich Stochastic Integrals of Multiplicity 8

This section is devoted to the following theorem.

Theorem 10 [16] (Sect. 2.37). *Suppose that* $\{\phi_j(x)\}_{j=0}^{\infty}$ *is a CONS of Legendre polynomials or trigonometric Fourier basis in* $L_2([t,T])$. *Then, for the*

iterated Stratonovich stochastic integral of eighth multiplicity

$$J^*[\psi^{(8)}]_{T,t} = \int\limits_t^{*T} \ldots \int\limits_t^{*t_2} d\mathbf{w}_{t_1}^{(i_1)} \ldots d\mathbf{w}_{t_8}^{(i_8)}$$

the following expansion:

$$J^*[\psi^{(8)}]_{T,t} = \underset{p\to\infty}{\text{l.i.m.}} \sum_{j_1,\ldots,j_8=0}^{p} C_{j_8\ldots j_1}\zeta_{j_1}^{(i_1)} \ldots \zeta_{j_8}^{(i_8)}$$

that converges in the mean-square sense is valid, where $i_1,\ldots,i_8 = 0,1,\ldots,m$,

$$C_{j_8\ldots j_1} = \int\limits_t^T \phi_{j_8}(t_8)\ldots \int\limits_t^{t_2} \phi_{j_1}(t_1)dt_1\ldots dt_8,$$

and

$$\zeta_j^{(i)} = \int\limits_t^T \phi_j(\tau)d\mathbf{w}_\tau^{(i)}$$

are independent standard Gaussian random variables for various i *or* j *(in the case* $i \neq 0$*),* $\mathbf{w}_\tau^{(0)} = \tau$.

Proof. To prove the theorem, we need to check the condition (63) (or the condition (64)) for the case $k = 8 > 2r$, where $r = 1,2,3$. Recall that the case $k = 2r$ is considered in [39]-[41] (also see [14], Sect. 3.2 or [16], Sect. 2.27.4). Under the conditions of Theorem 10, this means that $k = 8 = 2r$, where $r = 4$. The relations (65)–(77), (78) cover the case $k = 8$, $r = 1,2$ (see (63)). Thus, it remains to consider the case $k = 8$, $r = 3$. The case $k = 7$, $r = 3$ was considered in the previous section. Here we will focus on the differences between these two cases.

Since now $k = 8$, then along with inequalities (79)–(82), it is necessary to prove the following inequalities:

$$\left|\sum_{j_{g_1},j_{g_3},j_{g_5}=0}^{p} \left(C_{j_{d_3}j_{d_3-1}j_{d_3-2}j_{d_3-3}}(s,\tau)C_{j_{d_2}}(\theta,u)C_{j_{d_1}}(\rho,v)\right)\Big|_{j_{g_1}=j_{g_2},j_{g_3}=j_{g_4},j_{g_5}=j_{g_6}}\right| \le K, \tag{142}$$

$$\left|\sum_{j_{g_1},j_{g_3}.j_{g_5}=0}^{p}\left(C_{j_{d_3}j_{d_3-1}j_{d_3-2}}(s,\tau)C_{j_{d_2}j_{d_2-1}}(\theta,u)C_{j_{d_1}}(\rho,v)\right)\Bigg|_{j_{g_1}=j_{g_2},j_{g_3}=j_{g_4},j_{g_5}=j_{g_6}}\right|\le K, \tag{143}$$

$$\left|\sum_{j_{g_1},j_{g_3}.j_{g_5}=0}^{p}\left(C_{j_{d_3}j_{d_3-1}}(s,\tau)C_{j_{d_2}j_{d_2-1}}(\theta,u)C_{j_{d_1}j_{d_1-1}}(\rho,v)\right)\Bigg|_{j_{g_1}=j_{g_2},j_{g_3}=j_{g_4},j_{g_5}=j_{g_6}}\right|\le K, \tag{144}$$

where $p\in\mathbf{N}$, $t\le\tau<s\le T$, $t\le u<\theta\le T$, $t\le v<\rho\le T$, constant K does not depend on p,s,τ,θ,u,ρ,v (but only on t,T) and may differ from line to line; another notations are the same as in Sect. 3.3.

The inequalities (142)–(144) are proved using the same technique as inequalities (65)–(77) (see Sect. 3.3). Here we will only prove as an example the following special case of the inequality (144):

$$\left|\sum_{j_1,j_2,j_3=0}^{p}C_{j_2j_1}(s,\tau)C_{j_3j_1}(\theta,u)C_{j_2j_3}(\rho,v)\right|\le K. \tag{145}$$

Using the Cauchy–Bunyakovsky inequality as well as Fubini's Theorem, Parseval's equality and (66), we have

$$\left(\sum_{j_1,j_2,j_3=0}^{p}C_{j_2j_1}(s,\tau)C_{j_3j_1}(\theta,u)C_{j_2j_3}(\rho,v)\right)^2=$$

$$=\left(\sum_{j_2,j_3=0}^{p}C_{j_2j_3}(\rho,v)\sum_{j_1=0}^{p}C_{j_2j_1}(s,\tau)C_{j_3j_1}(\theta,u)\right)^2\le$$

$$\le\sum_{j_2,j_3=0}^{p}C^2_{j_2j_3}(\rho,v)\sum_{j_2,j_3=0}^{p}\left(\sum_{j_1=0}^{p}C_{j_2j_1}(s,\tau)C_{j_3j_1}(\theta,u)\right)^2\le$$

$$\le\sum_{j_2,j_3=0}^{\infty}C^2_{j_2j_3}(\rho,v)\sum_{j_2,j_3=0}^{\infty}\left(\sum_{j_1=0}^{p}C_{j_2j_1}(s,\tau)C_{j_3j_1}(\theta,u)\right)^2=$$

$$=\frac{(\rho-v)^2}{2}\sum_{j_2,j_3=0}^{\infty}\left(\sum_{j_1=0}^{p}\int\limits_{\tau}^{s}\phi_{j_2}(t_2)\int\limits_{\tau}^{t_2}\phi_{j_1}(t_1)dt_1dt_2\int\limits_{u}^{\theta}\phi_{j_3}(t_4)\int\limits_{u}^{t_4}\phi_{j_1}(t_3)dt_3dt_4\right)^2=$$

$$= \frac{(\rho - v)^2}{2} \sum_{j_2, j_3 = 0}^{\infty} \left(\int\limits_{\tau}^{s} \int\limits_{u}^{\theta} \phi_{j_2}(t_2) \phi_{j_3}(t_4) \times \right.$$

$$\left. \times \sum_{j_1=0}^{p} \int\limits_{\tau}^{t_2} \phi_{j_1}(t_1) dt_1 \int\limits_{u}^{t_4} \phi_{j_1}(t_3) dt_1 dt_3 dt_4 dt_2 \right)^2 =$$

$$= \frac{(\rho - v)^2}{2} \int\limits_{\tau}^{s} \int\limits_{u}^{\theta} \left(\sum_{j_1=0}^{p} C_{j_1}(t_2, \tau) C_{j_1}(t_4, u) \right)^2 dt_4 dt_2 \leq$$

$$\leq K_1^2 \frac{(\rho - v)^2}{2} (s - \tau)(\theta - u) \leq K_1^2 \frac{(T - t)^4}{2} = K.$$

The inequality (145) is proved.

The inequalities (79)–(82) for the case $k = 8$ are proved similarly to the inequalities (79)–(82) for the case $k = 7$ (see Sect. 3.3). There will be minor differences only when proving (79) for the case $k = 8$ (polynomial case). The above differences will be due to the fact that along with the two cases (129) the following third case:

$$\tau, s \in (t, T)$$

will now appear when proving (93), (94), (99).

Let us explain this in more detail. Cases 1 and 2 are obtained similarly to (100) and (101). Further, using Fubini's Theorem, we have

$$\int\limits_{t}^{T} h_7(t_7) \int\limits_{t}^{t_7} h_6(t_6) \int\limits_{t}^{t_6} h_5(t_5) \ldots \int\limits_{t}^{t_2} h_1(t_1) \int\limits_{t}^{t_1} h_8(t_8) dt_8 dt_1 \ldots dt_6 dt_7 =$$

$$= \int\limits_{t}^{T} h_7(t_7) \int\limits_{t}^{t_7} h_8(t_8) \int\limits_{t_8}^{t_7} h_1(t_1) \int\limits_{t_1}^{t_7} h_2(t_2) \ldots \int\limits_{t_5}^{t_7} h_6(t_6) dt_6 \ldots dt_1 dt_8 dt_7 =$$

$$= \int\limits_{t}^{T} h_7(t_7) \int\limits_{t}^{t_7} h_8(t_8) \int\limits_{t_8}^{t_7} h_6(t_6) \int\limits_{t_8}^{t_6} h_5(t_5) \ldots \int\limits_{t_8}^{t_2} h_1(t_1) dt_1 \ldots dt_6 dt_8 dt_7 =$$

$$= \int\limits_t^T h_7(s) \int\limits_t^s h_8(\tau) \left(\int\limits_\tau^s h_6(t_6) \int\limits_\tau^{t_6} h_5(t_5) \ldots \int\limits_\tau^{t_2} h_1(t_1) dt_1 \ldots dt_6 \right) d\tau ds, \quad (146)$$

where $h_1(\tau), \ldots, h_8(\tau) \in L_2([t,T])$. The equality (146) explains the occurrence of the third case.

Applying the technique that led to the estimates (134) and (140), we obtain for Case 3

$$\left(\sum_{j_1,j_2,j_3=0}^{p} C_{j_3 j_3 j_2 j_1 j_2 j_1}(s,\tau) \right)^2 \le \frac{K^2}{(1-z^2(\tau))^{3/8}} \stackrel{\mathsf{def}}{=} F(\tau) \quad (\text{for } (93)),$$

$$\left(\sum_{j_1,j_2,j_3=0}^{p} C_{j_3 j_3 j_1 j_2 j_2 j_1}(s,\tau) \right)^2 \le \frac{K^2}{(1-z^2(\tau))^{1/2}} \stackrel{\mathsf{def}}{=} F(\tau) \quad (\text{for } (94)),$$

$$\left(\sum_{j_1,j_2,j_3=0}^{p} C_{j_2 j_3 j_3 j_1 j_2 j_1}(s,\tau) \right)^2 \le \frac{K^2}{(1-z^2(\tau))^{3/8}} \stackrel{\mathsf{def}}{=} F(\tau) \quad (\text{for } (99)),$$

where constant K depends only on t, T and $F(\tau) \in L_1([t,T])$ (integrable majorant (see (64)). Thus, Theorem 10 is proved.

3.5 Special Cases for Which Theorem 7 is Valid Without the Condition (11)

In this section, we will collect some special cases for which the expansion (14) of the iterated Stratonovich stochastic integrals of multiplicity k ($k \in \mathbf{N}$) is true without the condition (11).

Case 1. *The numbers $i_1, \ldots, i_k$ are pairwise different ($i_1, \ldots, i_k = 1, \ldots, m$). It is also allowed that some of the numbers $i_1, \ldots, i_k$ are equal to zero (the remaining non-zero numbers from the set $\{i_1, \ldots, i_k\}$ are pairwise different).*

Consider the following representation for the multiple Wiener stochastic integral defined as in [1] but for the case of a multidimensional Wiener process (see [16], Sect. 1.11 for details).

Theorem 11 [16] (Theorem 1.23). *Suppose that* $\{\phi_j(x)\}_{j=0}^{\infty}$ *is an arbitrary CONS in* $L_2([t,T])$. *Then*

$$J'[\phi_{j_1}\ldots\phi_{j_k}]^{(i_1\ldots i_k)}=\prod_{l=1}^{k}\zeta_{j_l}^{(i_l)}+\sum_{r=1}^{[k/2]}(-1)^r\times$$

$$\times\sum_{\substack{(\{\{g_1,g_2\},\ldots,\{g_{2r-1},g_{2r}\}\},\{q_1,\ldots,q_{k-2r}\})\\ \{g_1,g_2,\ldots,g_{2r-1},g_{2r},q_1,\ldots,q_{k-2r}\}=\{1,2,\ldots,k\}}}\prod_{s=1}^{r}\mathbf{1}_{\{i_{g_{2s-1}}=\ i_{g_{2s}}\neq 0\}}\mathbf{1}_{\{j_{g_{2s-1}}=\ j_{g_{2s}}\}}\prod_{l=1}^{k-2r}\zeta_{j_{q_l}}^{(i_{q_l})}\qquad(147)$$

w. p. 1, *where* $i_1,\ldots,i_k=0,1,\ldots,m$, $[x]$ *is an integer part of a real number* x, $\prod\limits_{\emptyset}\stackrel{\mathsf{def}}{=}1$, $\sum\limits_{\emptyset}\stackrel{\mathsf{def}}{=}0$; *the sum in the second line of the formula* (147) *is the sum with respect to all possible partitions* (5); *another notations are the same as in Theorem* 1.

For Case 1, using Theorems 5, 8, and 11, we obtain the expansion (14).

Case 2. *The numbers* $i_1,\ldots,i_k$ *are such that* $i_1=\ldots=i_k=i\in\{1,\ldots,m\}$. *Moreover,* $\psi_1(\tau),\ldots,\psi_k(\tau)\equiv\psi(\tau)$, *where* $\psi(\tau)$ *is a continuous function on* $[t,T]$.

For Case 2, we have w. p. 1

$$J^*[\psi^{(k)}]_{T,t}^{(\overbrace{i\ \ldots\ i}^{k})}=\frac{1}{k!}\left(\int\limits_t^T\psi(\tau)d\mathbf{w}_\tau^{(i)}\right)^k=\frac{1}{k!}\left(\underset{p\to\infty}{\text{l.i.m.}}\sum_{j_1=0}^{p}C_{j_1}\zeta_{j_1}^{(i)}\right)^k=$$

$$=\underset{p\to\infty}{\text{l.i.m.}}\ \frac{1}{k!}\left(\sum_{j_1=0}^{p}C_{j_1}\zeta_{j_1}^{(i)}\right)^k=\underset{p\to\infty}{\text{l.i.m.}}\ \frac{1}{k!}\sum_{j_1,\ldots,j_k=0}^{p}C_{j_1}\ldots C_{j_k}\zeta_{j_1}^{(i)}\ldots\zeta_{j_k}^{(i)}=$$

$$=\underset{p\to\infty}{\text{l.i.m.}}\ \frac{1}{k!}\sum_{j_1,\ldots,j_k=0}^{p}\left(\sum_{(j_1,\ldots,j_k)}C_{j_k\ldots j_1}\right)\zeta_{j_1}^{(i)}\ldots\zeta_{j_k}^{(i)}=$$

$$=\underset{p\to\infty}{\text{l.i.m.}}\ \frac{1}{k!}\sum_{(j_1,\ldots,j_k)}\left(\sum_{j_1,\ldots,j_k=0}^{p}C_{j_k\ldots j_1}\zeta_{j_1}^{(i)}\ldots\zeta_{j_k}^{(i)}\right)=$$

$$=\underset{p\to\infty}{\text{l.i.m.}}\ \frac{1}{k!}\cdot k!\sum_{j_1,\ldots,j_k=0}^{p}C_{j_k\ldots j_1}\zeta_{j_1}^{(i)}\ldots\zeta_{j_k}^{(i)}=$$

$$= \underset{p\to\infty}{\text{l.i.m.}} \sum_{j_1,\ldots,j_k=0}^{p} C_{j_k\ldots j_1}\zeta_{j_1}^{(i)}\ldots\zeta_{j_k}^{(i)},$$

where notations as in Theorem 7,

$$\sum_{(j_1,\ldots,j_k)}$$

means the sum with respect to all possible permutations $(j_1,\ldots,j_k)$, and $\psi_1(\tau),\ldots,\psi_k(\tau)\equiv\psi(\tau)$, where $\psi(\tau)$ is a continuous function on $[t,T]$.

Case 3. *Among the numbers* $1,2,\ldots,k$ *there are numbers* $g_1,\ldots,g_r$ $(g_1+1<g_2,\ldots,g_{r-1}+1<g_r)$ *such that* $i_{g_1}=i_{g_1+1},\ldots,i_{g_r}=i_{g_r+1}$, *where* $r=1,2,\ldots,[k/2]$, $i_1,\ldots,i_k=1,\ldots,m$, *and the set*

$$\{i_1,\ldots,i_k\}\backslash\{i_{g_1},i_{g_1+1},\ldots,i_{g_r},i_{g_r+1}\}$$

consists of pairwise different elements, none of which is equal to $i_{g_1},i_{g_1+1},\ldots,$ i_{g_r},i_{g_r+1}. *Moreover,* $\psi_1(\tau),\ldots,\ \psi_k(\tau)\equiv 1$ *or* $\psi_1(\tau),\ldots,\psi_k(\tau)\equiv\psi(\tau)$, *where* $\psi(\tau)$ *is a continuous function on* $[t,T]$.

It is obvious that the left-hand side of condition (63) for Case 3 will decompose into a product of several expressions of the same type as the left-hand side of inequality (102). Thus, the condition (63) is satisfied for Case 3 and the expansion (14) is true for this case (see [16], Sect. 2.31 for details).

Consider the ordered set $(i_1,\ldots,i_k)$ that corresponds to Case 3. We have

$$(i_1,\ldots,i_{g_1-1},i_{g_1},i_{g_1},i_{g_1+2},\ldots,i_{g_r-1},i_{g_r},i_{g_r},i_{g_r+2},\ldots,i_k), \qquad (148)$$

where $r=1,2,\ \ldots,[k/2]$, $i_1,\ldots,i_k=1,\ldots,m$, and the set

$$\{i_1,\ldots,i_k\}\backslash\{i_{g_1},i_{g_1},\ldots,i_{g_r},i_{g_r}\}$$

consists of pairwise different elements, none of which is equal to $i_{g_1},\ldots,i_{g_r}$. Moreover, $i_{g_1},\ldots,i_{g_r}$ are pairwise different.

Consider some particular cases of (148).

Case 3.1:

$$(i_1, i_1, i_2, i_2, \ldots, i_r, i_r), \tag{149}$$

where $k = 2r$, $i_1, \ldots, i_r = 1, \ldots, m$ and $i_1, \ldots, i_r$ are pairwise different.

Case 3.2:

$$(i_1, i_1, i_3, i_4, i_4, i_6, \ldots, i_{3r-2}, i_{3r-2}, i_{3r}), \tag{150}$$

where $k = 3r$, $i_1, i_3, i_4, i_6, \ldots, i_{3r-2}, i_{3r} = 1, \ldots, m$ and $i_1, i_3, i_4, i_6, \ldots, i_{3r-2}, i_{3r}$ are pairwise different.

Also, we can consider a modification of (148) in the sense that $i_1, \ldots, i_k = 0, 1, \ldots, m$ in (148), i.e. some of the numbers $i_1, \ldots, i_k$ are equal to zero in (148).

In conclusion of this section, we will consider the expansion (14) of the iterated Stratonovich stochastic integral (1), which corresponds to (149) and (150). We have

$$J^*[\psi^{(2r)}]_{T,t}^{(i_1 i_1 i_2 i_2 \ldots i_r i_r)} =$$

$$= \underset{p\to\infty}{\text{l.i.m.}} \sum_{j_1, j_2, \ldots, j_{2r}=0}^{p} C_{j_{2r}\ldots j_2 j_1} \zeta_{j_1}^{(i_1)} \zeta_{j_2}^{(i_1)} \zeta_{j_3}^{(i_2)} \zeta_{j_4}^{(i_2)} \ldots \zeta_{j_{2r-1}}^{(i_{2r})} \zeta_{j_{2r}}^{(i_{2r})},$$

$$J^*[\psi^{(3r)}]_{T,t}^{(i_1 i_1 i_3 i_4 i_4 i_6 \ldots i_{3r-2} i_{3r-2} i_{3r})} =$$

$$= \underset{p\to\infty}{\text{l.i.m.}} \sum_{j_1, j_2, \ldots, j_{3r}=0}^{p} C_{j_{3r}\ldots j_2 j_1} \zeta_{j_1}^{(i_1)} \zeta_{j_2}^{(i_1)} \zeta_{j_3}^{(i_3)} \zeta_{j_4}^{(i_4)} \zeta_{j_5}^{(i_4)} \zeta_{j_6}^{(i_6)} \ldots \zeta_{j_{3r-2}}^{(i_{3r-2})} \zeta_{j_{3r-1}}^{(i_{3r-2})} \zeta_{j_{3r}}^{(i_{3r})},$$

where notations as in Theorem 7 and $\psi_1(\tau), \ldots, \psi_{3r}(\tau) \equiv \psi(\tau)$, where $\psi(\tau)$ is a continuous function on $[t, T]$ (in particular $\psi(\tau) \equiv 1$).

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